SETK 4643

Thermodynamics

Chapter 6: The Second Law Of Thermodynamics

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6.1 Introduction to the Second Law

- The second law of thermodynamics states that processes occur in a certain direction and not in the reverse direction.
- A process will not occur unless it satisfies both the first and the second laws of thermodynamics.
- Satisfying the first law alone does not ensure the process will actually take place

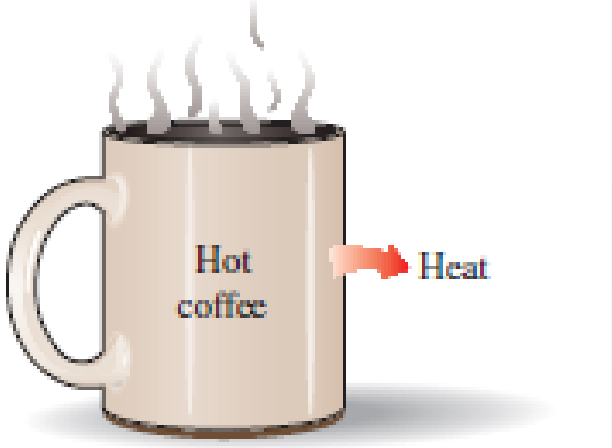


FIGURE 6–1
A cup of hot coffee does not get hotter in a cooler room.

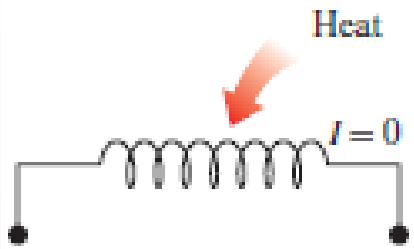


FIGURE 6–2
Transferring heat to a wire will not generate electricity.

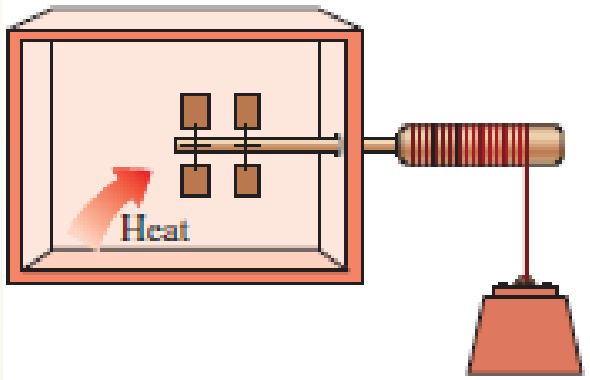


FIGURE 6–3
Transferring heat to a paddle wheel will not cause it to rotate.

6.1 Introduction to the Second Law (2)

- The second law may be used to identify the **direction** of processes.
- The second law also asserts that energy has **quality** as well as **quantity**.
  - The first law is concerned with the quantity of energy and the transformations of energy from one form to another with no regard to its quality.
  - The second law provides the necessary means to determine the quality as well as the degree of degradation of energy during a process.
- The second law of thermodynamics is also used in determining the **theoretical limits** for the performance of commonly used engineering systems, such as heat engines and refrigerators, as well as predicting the degree of completion of chemical reactions.

FIGURE 6–4
Processes occur in a certain direction, and not in the reverse direction.

FIGURE 6–5
A process must satisfy both the first and second laws of thermodynamics to proceed.

6.2 Thermal Energy Reservoirs

- Thermal energy reservoir, or just a reservoir is a hypothetical body with a relatively large thermal energy capacity (mass x specific heat) that can supply or absorb finite amounts of heat without undergoing any change in temperature.
  - In practice, oceans, lakes, rivers and the atmospheric air can be modeled accurately as thermal energy reservoirs
- A reservoir that supplies energy in the form of heat is called a **source**.
- A reservoir that absorbs energy in the form of heat is called a **sink**.
- Thermal energy reservoirs are often referred to as **heat reservoirs** since they supply or absorb energy in the form of heat.

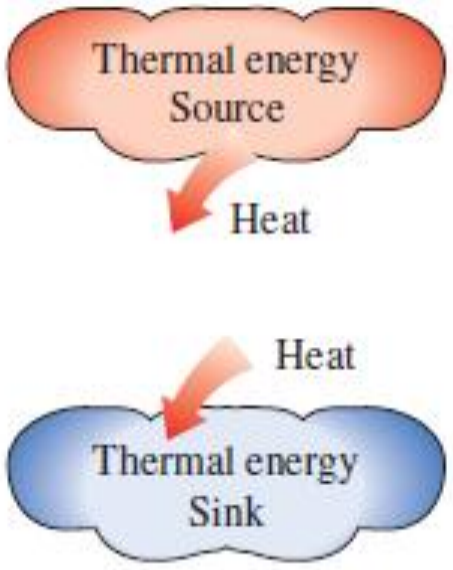


FIGURE 6–7
A source supplies energy in the form of heat, and a sink absorbs it.

6.3 Heat Engines

- Work can easily be converted to other forms of energy, but converting other forms of energy to work is not that easy.
- Heat engines are devices that convert heat to work.
- Heat engines differ considerably from one another, but all can be characterized by the following:
  1. They receive heat from a high-temperature source (solar energy, oil furnace, nuclear reactor, etc.).
  2. They convert part of this heat to work (usually in the form of a rotating shaft).
  3. They reject the remaining waste heat to a low-temperature sink (the atmosphere, rivers, etc.).
  4. They operate on a cycle.

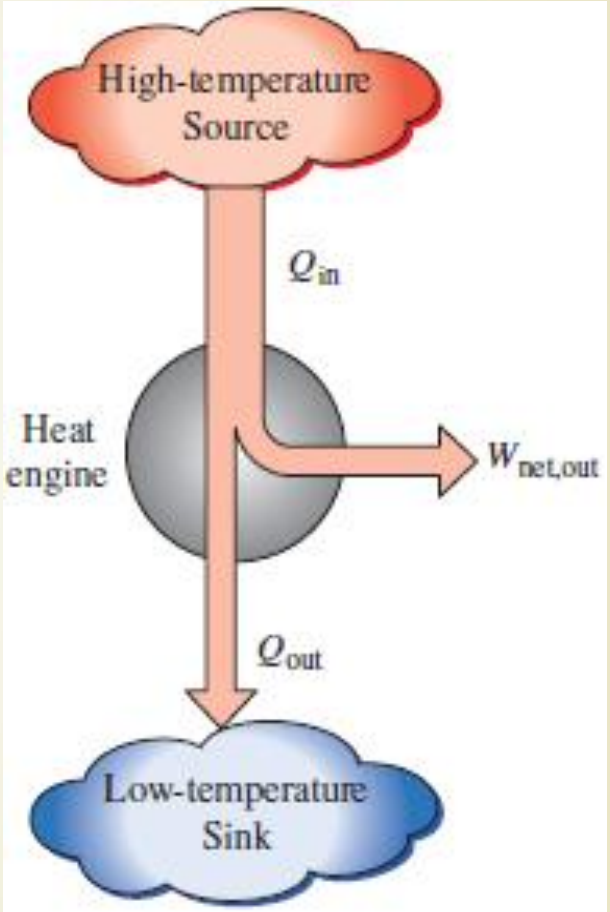
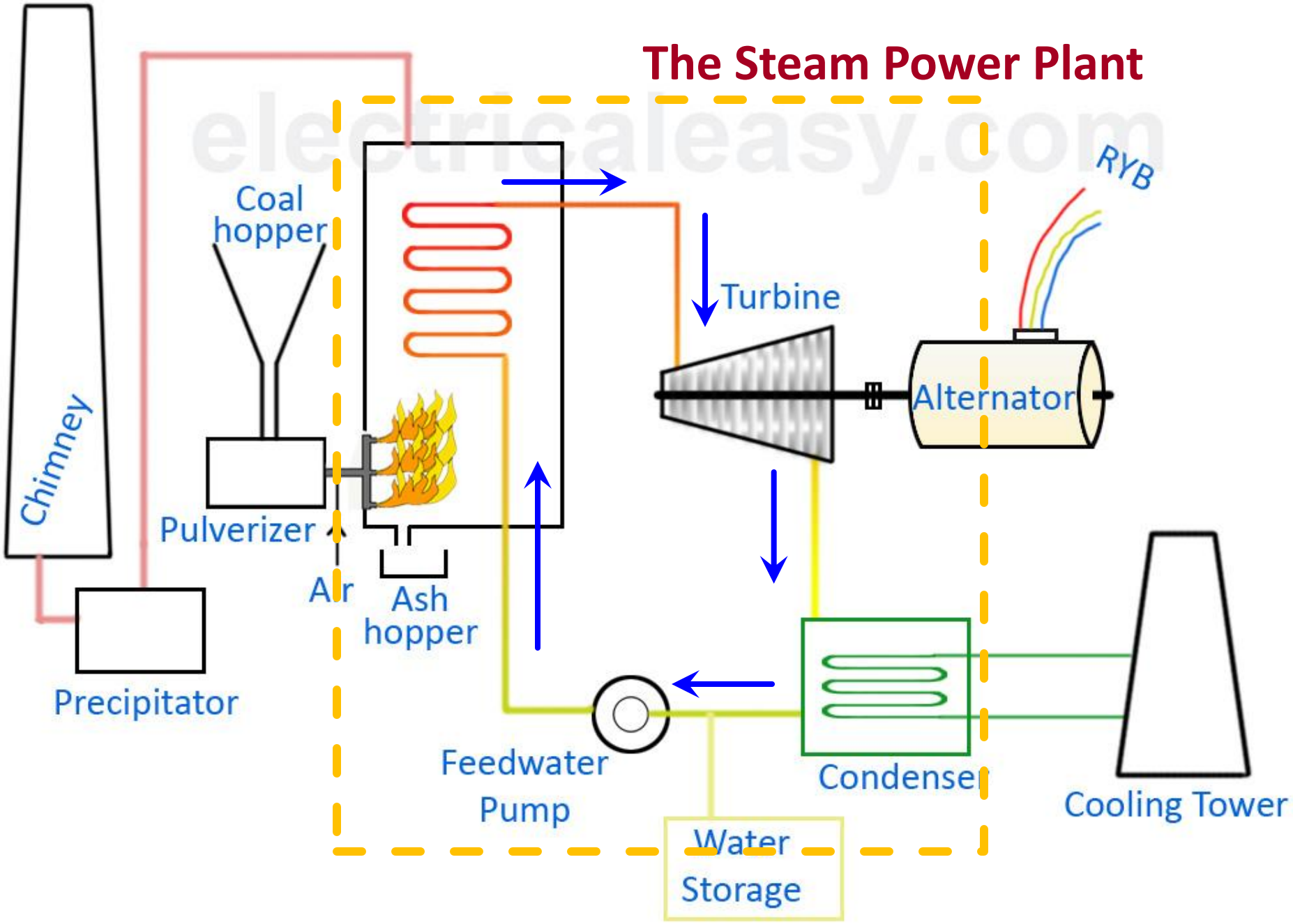


FIGURE 6–9
Part of the heat received by a heat engine is converted to work, while the rest is rejected to a sink.

6.3 Heat Engines (2)

- Heat engines and other cyclic devices usually involve a fluid to and from which heat and work is transferred while undergoing a cycle.
  - This fluid is called the working fluid.
- Engines that involve internal combustion such as gas turbines and car engines fall into the category of heat engines.
  - They operate in a mechanical cycle but not in a thermodynamic cycle since the working fluid (the combustion gases) does not undergo a complete cycle.
  - Instead of being cooled to the initial temperature, the exhaust gases are purged and replaced by fresh air-and-fuel mixture at the end of the cycle.
- The work-producing device that best fits into the definition of a heat engine is the steam power plant, which is an **external-combustion engine**.
  - combustion takes place outside the engine, and
  - the thermal energy released during this process is transferred to the steam as heat.

The Steam Power Plant



6.3 Heat Engines (3)

A steam power plant

- The various quantities shown are:
  - Q_{in} = amount of heat supplied to steam in boiler from a high-temperature source
  - Q_{out} = amount of heat rejected from steam in condenser to a low-temperature sink (the atmosphere, a river, etc.)
  - W_{out} = amount of work delivered by steam as it expands in turbine
  - W_{in} = amount of work required to compress water to boiler pressure

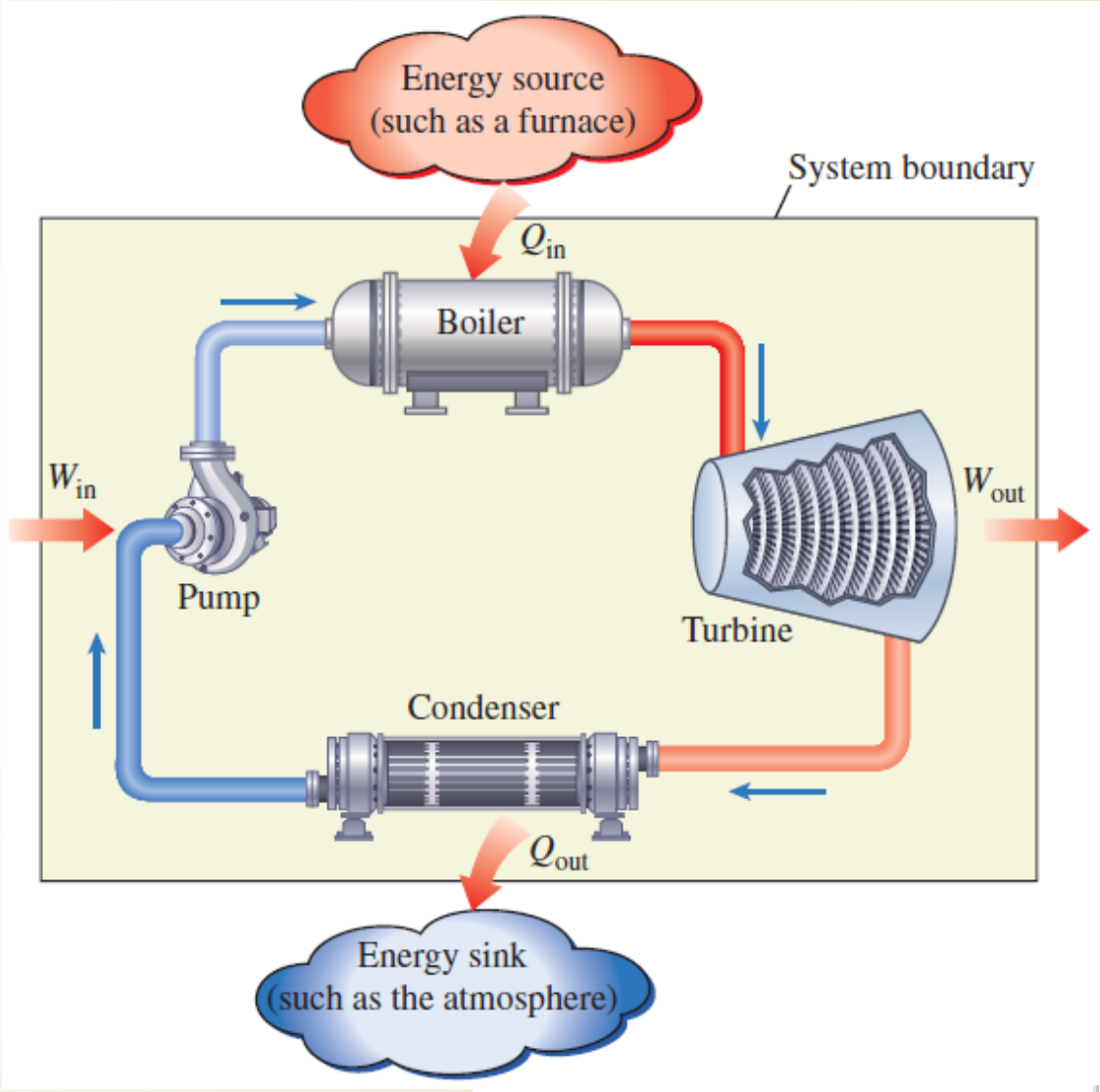


FIGURE 6–10
Schematic of a steam power plant.

6.3 Heat Engines (4)

- The net work output of the power plant is

$$W_{net,out} = W_{out} - W_{in} \qquad (kJ)$$

- The four components of the steam power plant involve mass flow in and out, and therefore should be treated as open systems.
- But these components, together with the connecting pipes, however, always contain the same fluid
  - No mass enters or leaves this combination system.
  - Thus, it can be analyzed as a closed system.
- For a closed system undergoing a cycle, the change in internal energy $\Delta U = 0$.

$$W_{net,out} = Q_{in} - Q_{out} \qquad (kJ)$$

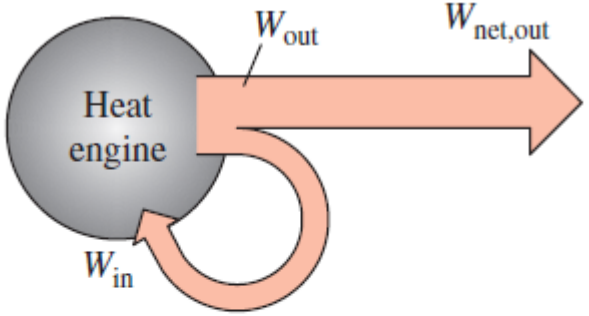


FIGURE 6–11
A portion of the work output of a heat engine is consumed internally to maintain continuous operation.

6.3 Heat Engines (5)

Thermal Efficiency

- Q_{out} represents the magnitude of the energy wasted in order to complete the cycle.
- Q_{out} is **never zero**;
  - thus, the net work output of a heat engine is always less than the amount of heat input.
- Only part of the heat transferred to the heat engine is converted to work.
- **Thermal efficiency**, η_{th} is a measure of the performance of a heat engine
  - the fraction of the heat input to the heat engine that is converted to net work output.

$$\text{Thermal efficiency} = \frac{\text{Net work output}}{\text{Total heat input}}$$

or

$$\eta_{th} = \frac{W_{net,out}}{Q_{in}}$$

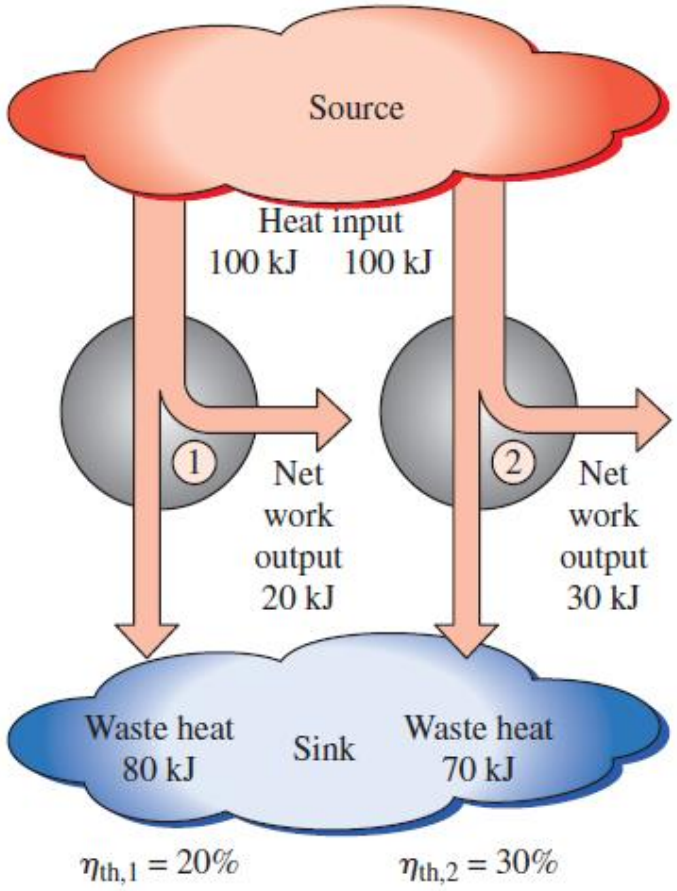


FIGURE 6–12
Some heat engines perform better than others (convert more of the heat they receive to work).

6.3 Heat Engines (6)

- It can also be expressed as

$$\eta_{th} = 1 - \frac{Q_{out}}{Q_{in}}$$ since $$W_{net,out} = Q_{in} - Q_{out} \quad \text{(kJ)}$$

- Cyclic devices of practical interest such as heat engines, refrigerators, and heat pumps operate between a high-temperature reservoir at T_H and a low-temperature reservoir at T_L.
  - Q_H = magnitude of heat transfer between the cyclic device and the high temperature medium at temperature T_H.
  - Q_L = magnitude of heat transfer between the cyclic device and the low- temperature medium at temperature T_L.
- Notice that both Q_L and Q_H are defined as magnitudes and therefore are positive quantities.
  - The direction of Q_H and Q_L is easily determined by inspection.

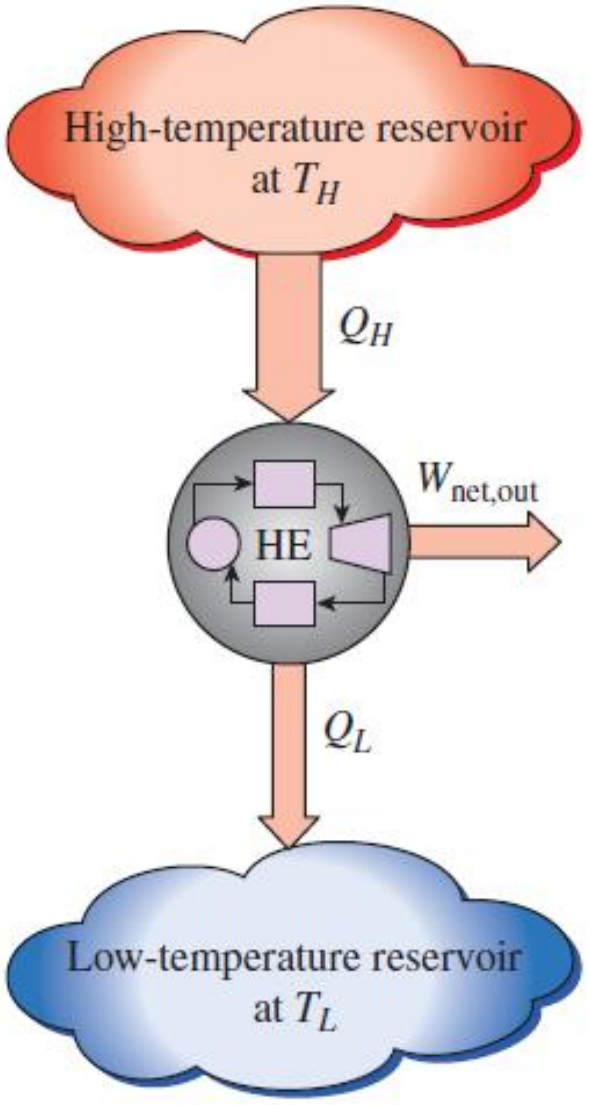


FIGURE 6–13
Schematic of a heat engine.

6.3 Heat Engines (7)

- The net work output and thermal efficiency relations for any heat engine can also be expressed as

$$W_{\text{net,out}} = Q_H - Q_L$$

and

$$\eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_H} \quad \text{or} \quad \eta_{\text{th}} = 1 - \frac{Q_L}{Q_H}$$

- The thermal efficiency of a heat engine is always less than unity since both Q_L and Q_H are defined as positive quantities.
- Thermal efficiency is a measure of how efficiently a heat engine converts the heat it receives to work.
  - increased efficiency means less fuel consumption and thus lower fuel bills and less pollution.
  - spark-ignition automobile engines, $\eta_{\text{th}} = 25\%$
  - diesel (compression ignition) engines, $\eta_{\text{th}} = 40\%$
  - large power plants (combined gas-steam), $\eta_{\text{th}} = 60\%$

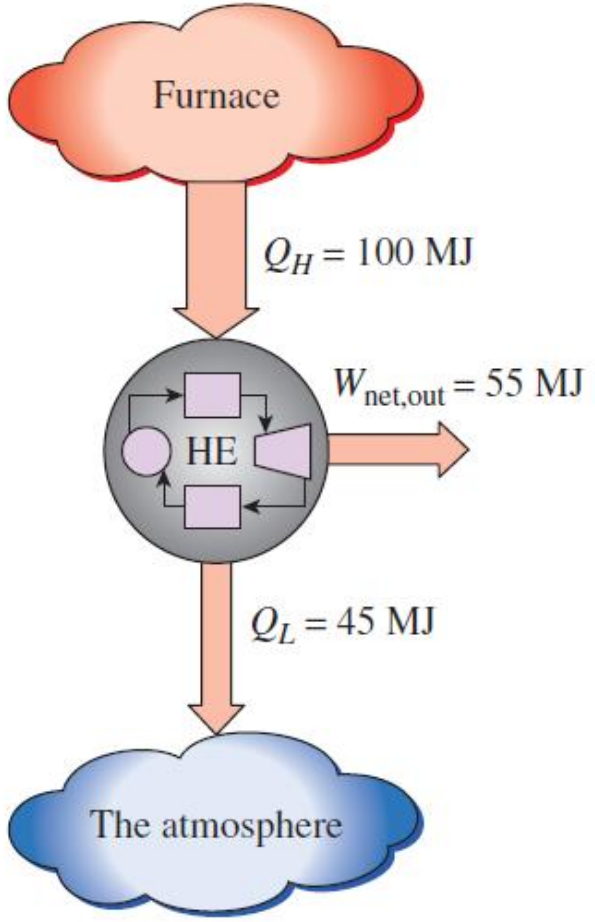


FIGURE 6–14
Even the most efficient heat engines reject almost one-half of the energy they receive as waste heat.

EXAMPLE 6–1 Net Power Production of a Heat Engine

Heat is transferred to a heat engine from a furnace at a rate of 80 MW. If the rate of waste heat rejection to a nearby river is 50 MW, determine the net power output and the thermal efficiency for this heat engine.

SOLUTION The rates of heat transfer to and from a heat engine are given. The net power output and the thermal efficiency are to be determined.

Assumptions Heat losses through the pipes and other components are negligible.

Analysis A schematic of the heat engine is given in Fig. 6–16. The furnace serves as the high-temperature reservoir for this heat engine and the river as the low-temperature reservoir. The given quantities can be expressed as

$$\dot{Q}_H = 80 \text{ MW} \quad \text{and} \quad \dot{Q}_L = 50 \text{ MW}$$

The net power output of this heat engine is

$$\dot{W}_{\text{net,out}} = \dot{Q}_H - \dot{Q}_L = (80 - 50) \text{ MW} = \mathbf{30 \text{ MW}}$$

Then the thermal efficiency is easily determined to be

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{net,out}}}{\dot{Q}_H} = \frac{30 \text{ MW}}{80 \text{ MW}} = \mathbf{0.375} \text{ (or 37.5 percent)}$$

Discussion Note that the heat engine converts 37.5 percent of the heat it receives to work.

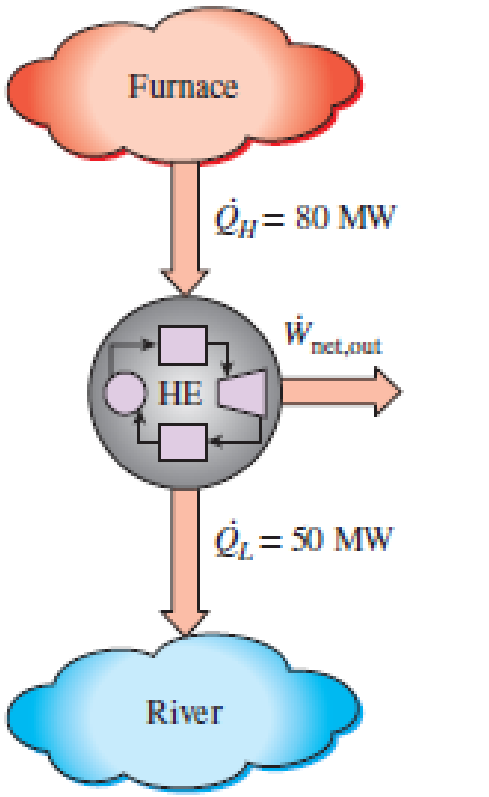


FIGURE 6–16
Schematic for Example 6–1.

6.3 Heat Engines (8)

Can we save Q_{out}?

- In a steam power plant, the condenser is the device where large quantities of waste heat is rejected to rivers, lakes, or the atmosphere.
- Can we not just take the condenser out of the plant and save all that waste energy?
- The answer is a firm **no** since without a heat rejection process in a condenser, the cycle cannot be completed.
- Every heat engine **must** waste some energy by transferring it to a low-temperature reservoir in order to complete the cycle, even under idealized conditions.
- This is called **energy waste**

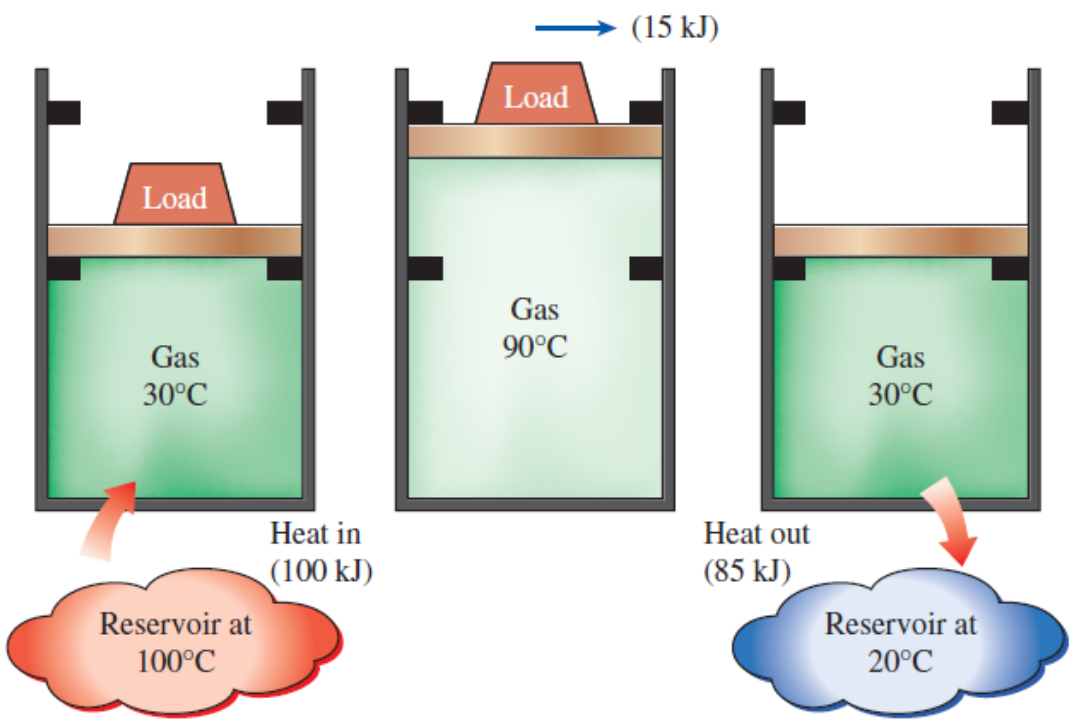


FIGURE 6–15
A heat-engine cycle cannot be completed without rejecting some heat to a low-temperature sink.

6.3 Heat Engines (9)

The Second Law of Thermodynamics: Kelvin-Planck Statement

- Kelvin-Planck statement of the second law of thermodynamics is expressed as follows:

It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce a net amount of work.

- The statement can also be expressed as

No heat engine can have a thermal efficiency of 100 percent

or

For a power plant to operate, the working fluid must exchange heat with the environment as well as the furnace.

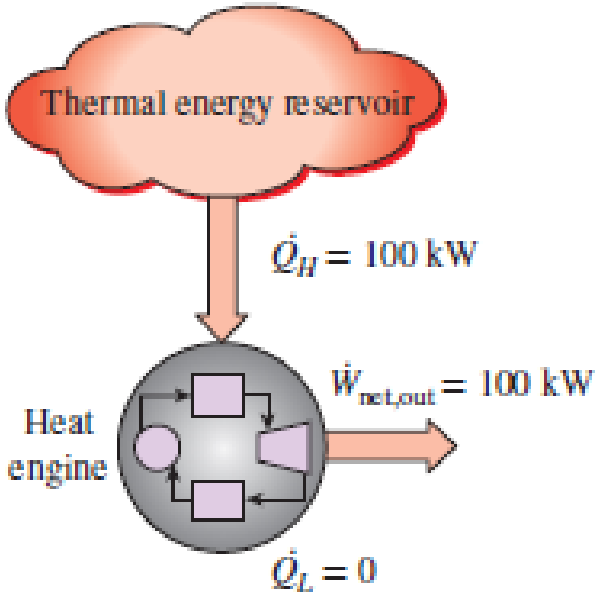


FIGURE 6–18
A heat engine that violates the Kelvin–Planck statement of the second law.

Exercise 1

6–15 A steam power plant receives heat from a furnace at a rate of 280 GJ/h. Heat losses to the surrounding air from the steam as it passes through the pipes and other components are estimated to be about 8 GJ/h. If the waste heat is transferred to the cooling water at a rate of 145 GJ/h, determine (*a*) net power output and (*b*) the thermal efficiency of this power plant. *Answers:* (*a*) 35.3 MW, (*b*) 45.4 percent

Exercise 2

6–19 A 600-MW steam power plant, which is cooled by a nearby river, has a thermal efficiency of 40 percent. Determine the rate of heat transfer to the river water. Will the actual heat transfer rate be higher or lower than this value? Why?

6.4 Refrigerators and heat Pumps

- Refrigerators are cyclic devices which allow the transfer of heat from a low-temperature medium to a high-temperature medium.
- Refrigerators, like heat engines, are cyclic devices.
- The working fluid used in the refrigeration cycle is called a **refrigerant**.
- The most frequently used refrigeration cycle is the **vapor-compression refrigeration cycle**, which involves four main components:
  - a compressor, a condenser, an expansion valve, and an evaporator.

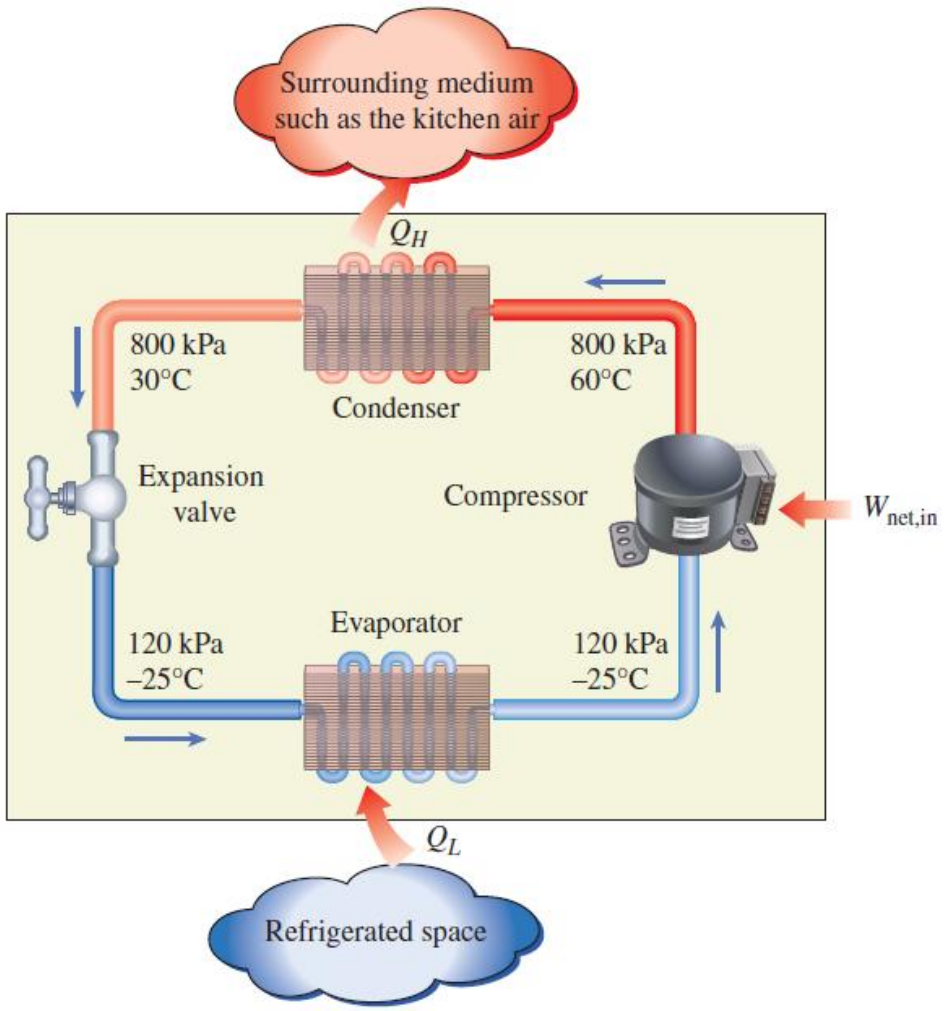
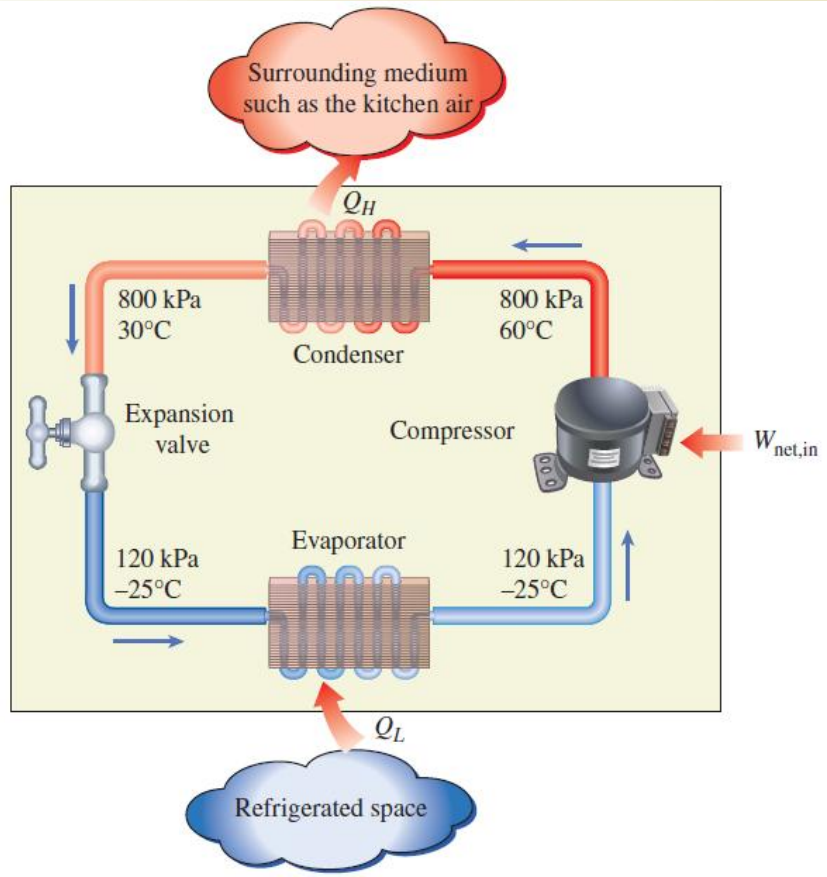


FIGURE 6–19
Basic components of a refrigeration system and typical operating conditions.

6.4 Refrigerators and heat Pumps (2)

1. The refrigerant enters the compressor as a vapor and is compressed to the condenser pressure.
2. It leaves the compressor at a relatively high temperature and cools down and condenses as it flows through the coils of the condenser by rejecting heat to the surrounding medium.
3. It then enters a capillary tube where its pressure and temperature drop drastically due to the throttling effect.
4. The low-temperature refrigerant then enters the evaporator, where it evaporates by absorbing heat from the refrigerated space.
5. The cycle is completed as the refrigerant leaves the evaporator and reenters the compressor.



In a household refrigerator, the freezer compartment where heat is absorbed by the refrigerant serves as the evaporator, and the coils usually behind the refrigerator where heat is dissipated to the kitchen air serve as the condenser.

6.4 Refrigerators and heat Pumps (3)

- A refrigerator is shown schematically in Fig. 6–20
  - Q_L is the magnitude of heat removed from the refrigerated space at temperature T_L,
  - Q_H is the magnitude of the heat rejected to the warm environment at temperature T_H, and
  - $W_{net,in}$ is the net work input to the refrigerator.
  - Q_L and Q_H represent magnitudes and thus are positive quantities.
- Air conditioners are basically refrigerators whose refrigerated space is a room or a building instead of the food compartment.
  - A window air-conditioning unit cools a room by absorbing heat from the room air and discharging it to the outside.

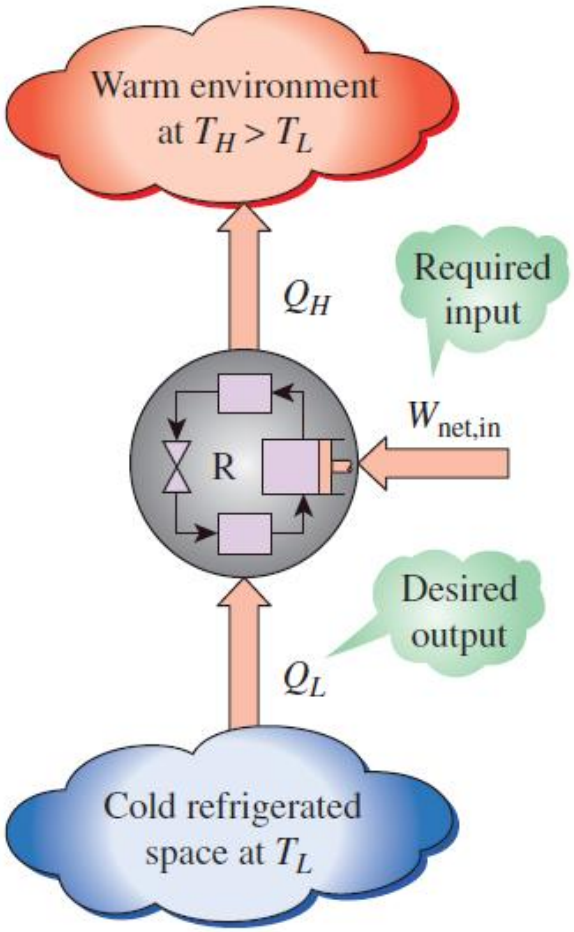


FIGURE 6–20
The objective of a refrigerator is to remove Q_L from the cooled space.

6.4 Refrigerators and heat Pumps (4)

Coefficient of performance

- The efficiency of a refrigerator is expressed in terms of the coefficient of performance (COP), denoted by COP_R.
  - The objective of a refrigerator is to remove heat (Q_L) from the refrigerated space.
  - To accomplish this objective , it requires a work input of $W_{net,in}$.
- Then the COP of a refrigerator can be expressed as

$$COP_R = \frac{\text{Desired output}}{\text{Required input}} = \frac{Q_L}{W_{net,in}}$$

- The conservation of energy principle for a cyclic device requires that $W_{net,in} = Q_H - Q_L \quad \text{(kJ)}$
- Then the COP relation becomes

$$COP_R = \frac{Q_L}{Q_H - Q_L} = \frac{1}{Q_H/Q_L - 1}$$

COP_R can be > 1

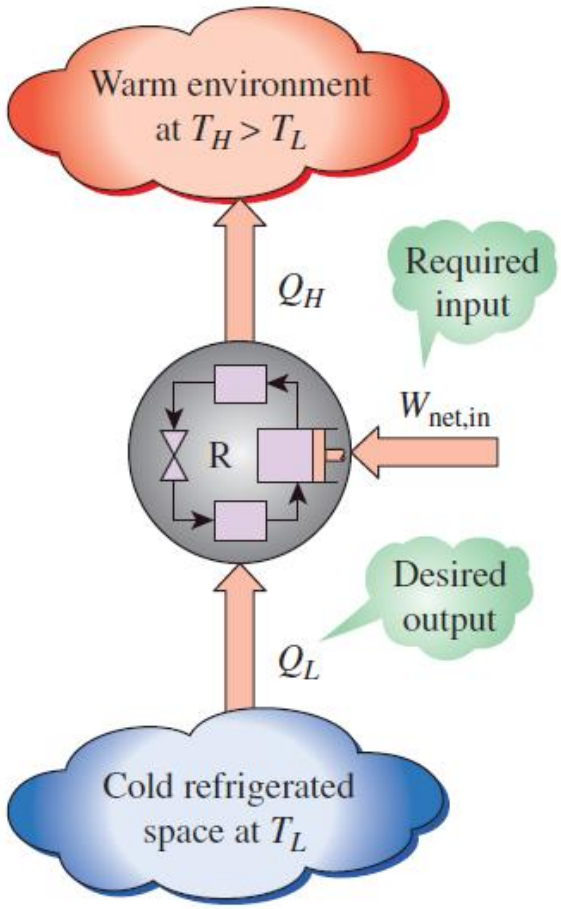


FIGURE 6–20
The objective of a refrigerator is to remove Q_L from the cooled space.

6.4 Refrigerators and heat Pumps (5)

Heat Pumps

- Another device that transfers heat from a low-temperature medium to a high temperature one is the **heat pump**
- Heat pumps are cyclic devices which operate on the refrigeration cycle and discharge energy to a heated space to maintain the heated space at a high temperature.
- Refrigerators and heat pumps operate on the same cycle but differ in their objectives.
  - refrigerator maintains the refrigerated space at a low temp. by removing heat from it
  - heat pump maintains a heated space at a high temp. by absorbing heat from a low-temp. source.
  - air-conditioning unit can be used as a heat pump in winter by installing it backwards

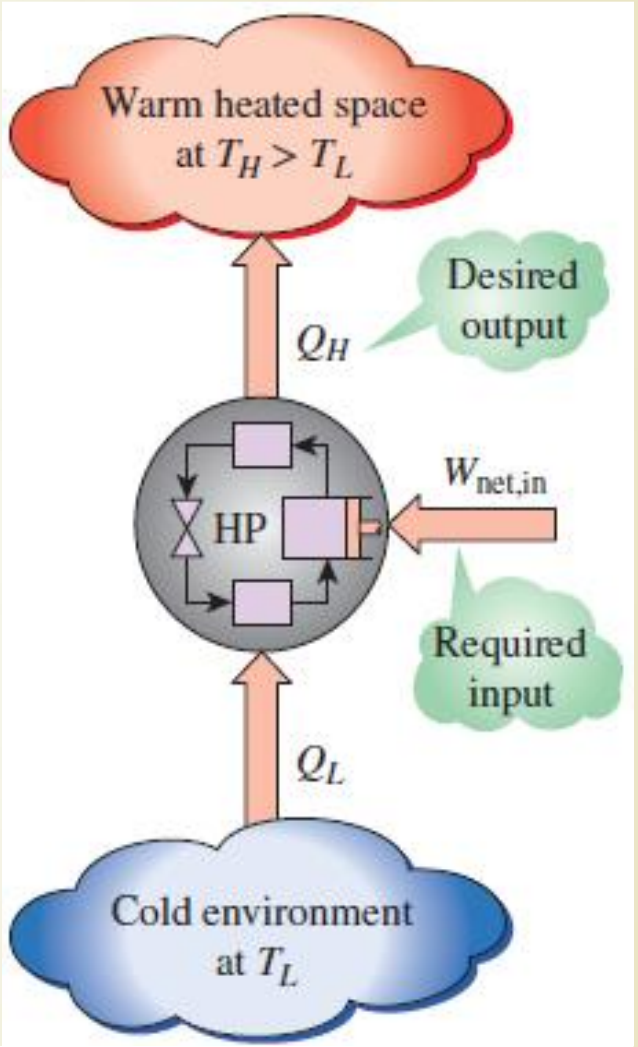


FIGURE 6–21
The objective of a heat pump is to supply heat Q_H into the warmer space.

6.4 Refrigerators and heat Pumps (6)

- The measure of performance of a heat pump is also expressed in terms of the coefficient of performance COP_{HP}, defined as

$$COP_{HP} = \frac{\text{Desired output}}{\text{Required input}} = \frac{Q_H}{W_{net,in}}$$

which can also be expressed as

$$COP_{HP} = \frac{Q_H}{Q_H - Q_L} = \frac{1}{1 - Q_L/Q_H}$$

- A comparison with previous equation reveals that

$$COP_{HP} = COP_R + 1$$ (for fixed values of Q_L and Q_H)

- This relation implies that the COP_{HP} is always greater than unity since COP_R is a positive quantity.

The heat source	COP_{HP}
cold outside air	2 – 3
geothermal	6

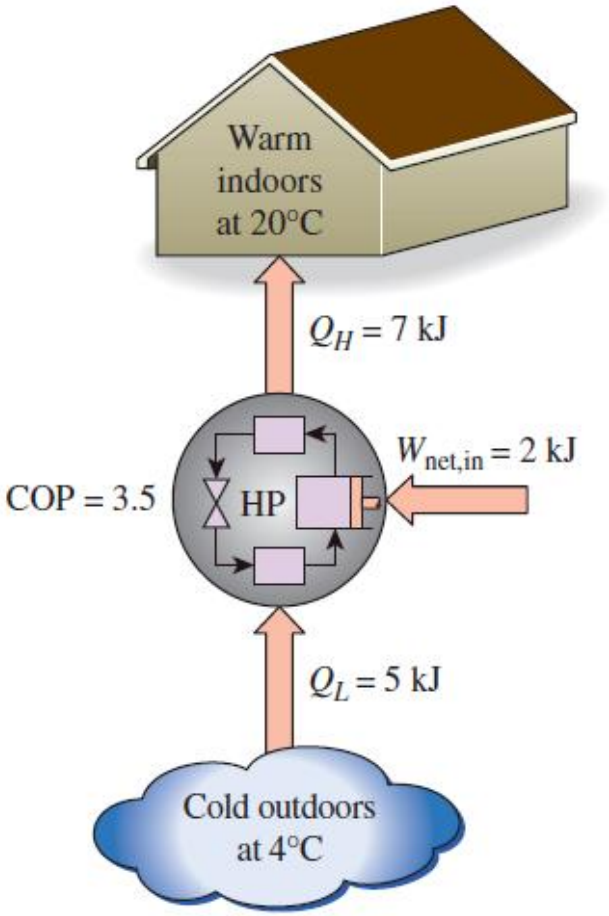


FIGURE 6–22
The work supplied to a heat pump is used to extract energy from the cold outdoors and carry it into the warm indoors.

EXAMPLE 6–3 Analysis of a Household Refrigerator

A household refrigerator with a COP of 1.2 removes heat from the refrigerated space at a rate of 60 kJ/min (Fig. 6-23). Determine (*a*) the electric power consumed by the refrigerator and (*b*) the rate of heat transfer to the kitchen air.

SOLUTION The COP and the refrigeration rate of a refrigerator are given. The power consumption and the rate of heat rejection are to be determined.

Assumptions The refrigerator operates steadily.

Analysis (*a*) Using the definition of the coefficient of performance, the power input to the refrigerator is determined to be

$$\dot{W}_{net,in} = \frac{\dot{Q}_L}{COP_R} = \frac{60 \text{ kJ/min}}{1.2} = 50 \text{ kJ/min} = \mathbf{0.833 \text{ kW}}$$

(*b*) The heat transfer rate to the kitchen air is determined from the energy balance,

$$\dot{Q}_H = \dot{Q}_L + \dot{W}_{net,in} = 60 + 50 = \mathbf{110 \text{ kJ/min}}$$

Discussion Notice that both the energy removed from the refrigerated space as heat and the energy supplied to the refrigerator as electrical work eventually show up in the room air and become part of the internal energy of the air. This demonstrates that energy can change from one form to another and can move from one place to another, but it is never destroyed during a process.

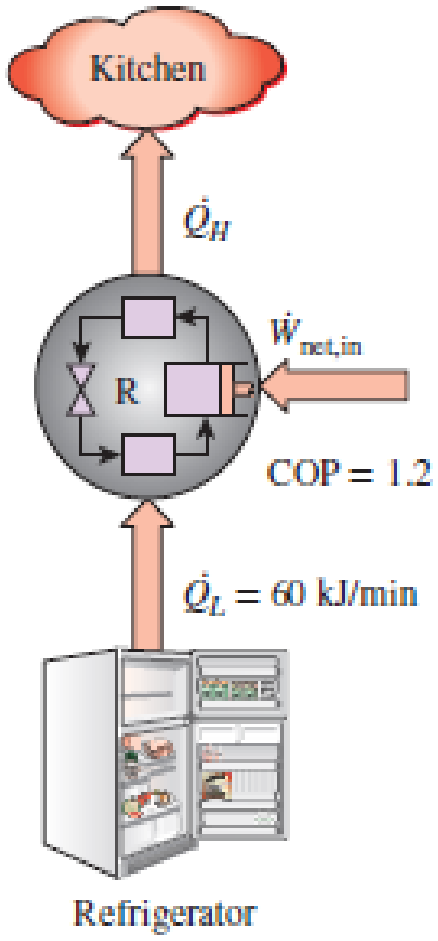


FIGURE 6–23
Schematic for Example 6–3.

EXAMPLE 6–4 Heating a House with a Heat Pump

A heat pump is used to meet the heating requirements of a house and maintain it at 20°C. On a day when the outdoor air temperature drops to –2°C, the house is estimated to lose heat at a rate of 80,000 kJ/h. If the heat pump under these conditions has a COP of 2.5, determine (*a*) the power consumed by the heat pump and (*b*) the rate at which heat is absorbed from the cold outdoor air.

SOLUTION The COP of a heat pump is given. The power consumption and the rate of heat absorption are to be determined.

Assumptions Steady operating conditions exist.

Analysis (*a*) The power consumed by this heat pump, shown in Fig. 6–24, is determined from the definition of the coefficient of performance to be

$$\dot{W}_{net,in} = \frac{\dot{Q}_H}{COP_{HP}} = \frac{80{,}000 \text{ kJ/h}}{2.5} = \mathbf{32{,}000 \text{ kJ/h}} \text{ (or 8.9 kW)}$$

(*b*) The house is losing heat at a rate of 80,000 kJ/h. If the house is to be maintained at a constant temperature of 20°C, the heat pump must deliver heat to the house at the same rate, that is, at a rate of 80,000 kJ/h. Then the rate of heat transfer from the outdoor becomes

$$\dot{Q}_L = \dot{Q}_H - \dot{W}_{net,in} = (80{,}000 - 32{,}000) \text{ kJ/h} = \mathbf{48{,}000 \text{ kJ/h}}$$

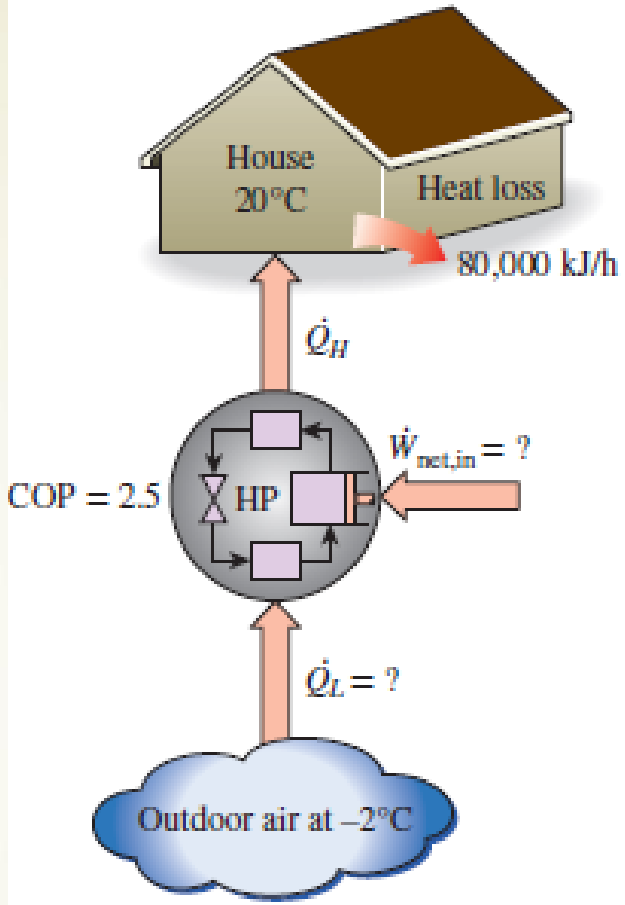


FIGURE 6–24
Schematic for Example 6–4.

Discussion Note that 48,000 of the 80,000 kJ/h heat delivered to the house is actually extracted from the cold outdoor air. Therefore, we are paying only for the 32,000 kJ/h energy that is supplied as electrical work to the heat pump. If we were to use an electric resistance heater instead, we would have to supply the entire 80,000 kJ/h to the resistance heater as electric energy. This would mean a heating bill that is 2.5 times higher. This explains the popularity of heat pumps as heating systems and why they are preferred to simple electric resistance heaters despite their considerably higher initial cost.

6.4 Refrigerators and heat Pumps (7)

The Second Law of Thermodynamics: Clausius Statement

- the Clausius statement of the second law of thermodynamics , which is related to refrigerators or heat pumps, is expressed as follows:

It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a lower-temperature body to a higher-temperature body.

- It states that a refrigerator cannot operate unless its compressor is driven by an external power source.
  - the net effect on the surroundings involves the consumption of some energy in the form of work, in addition to the transfer of heat from a colder body to a warmer one.
- To date, no experiment has been conducted that contradicts the second law, and this should be taken as sufficient proof of its validity.

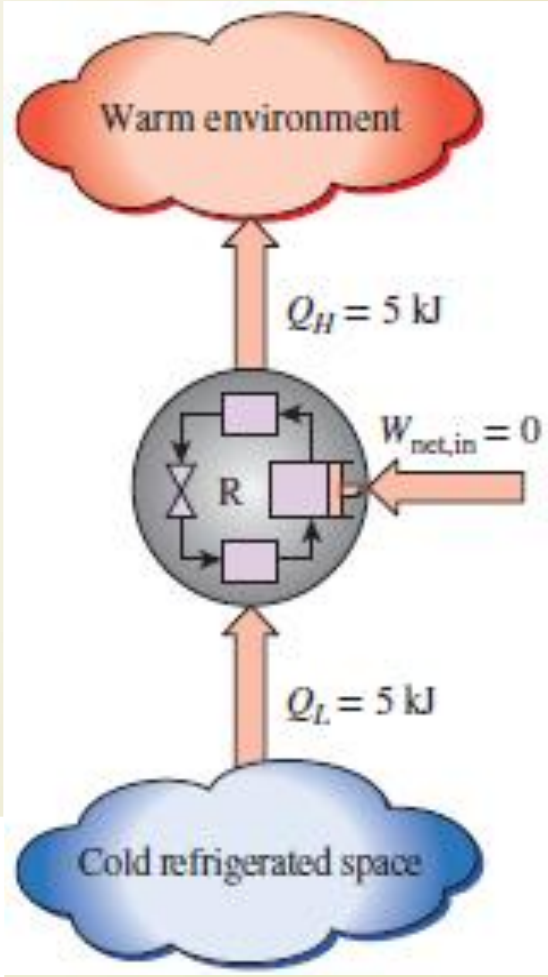


FIGURE 6–25
A refrigerator that violates the Clausius statement of the second law.

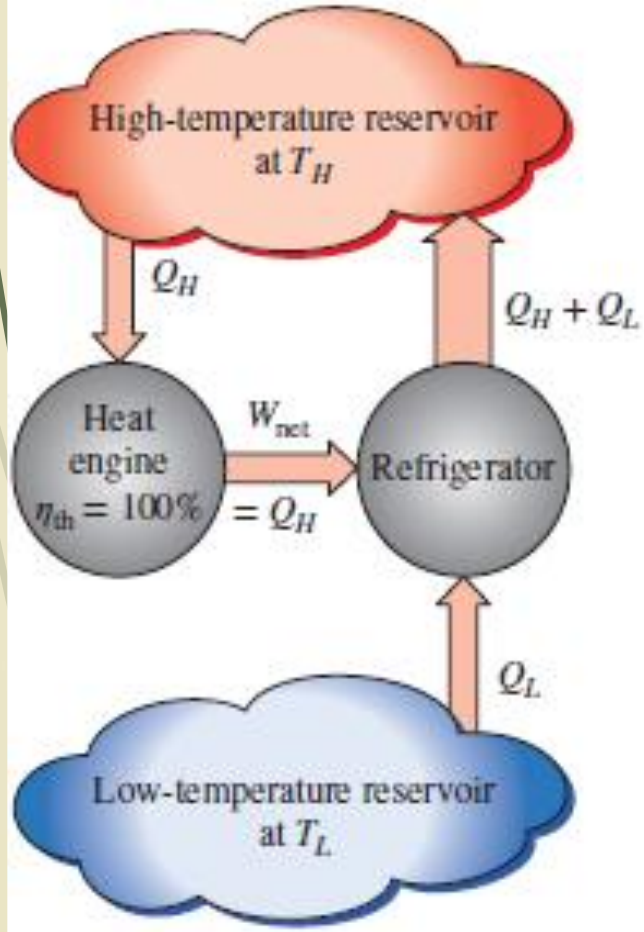
6.4 Refrigerators and heat Pumps (8)

- Both the Kelvin-Planck and the Clausius statements of the 2nd law are negative statements.
  - A negative statement cannot be proved.
  - The 2nd law of thermodynamics is based on experimental observations.
- The Kelvin–Planck and the Clausius statements are equivalent in their consequences, and either statement can be used as the expression of the second law of thermodynamics.
- Any device that violates the Kelvin–Planck statement also violates the Clausius statement, and vice versa.

6.4 Refrigerators and heat Pumps (9)

Equivalence of the Two Statements

- Consider the heat engine-refrigerator combination shown



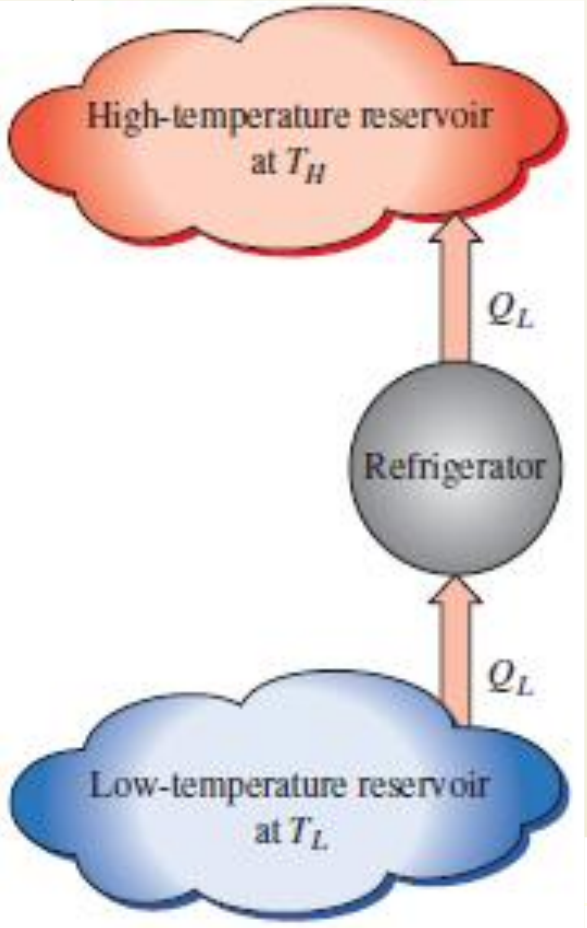
(*a*) A refrigerator that is powered by a 100 percent efficient heat engine

- The heat engine is assumed to have, in violation of the Kelvin-Planck statement by converting all the heat Q_H it receives to work *W, i.e.* $W = Q_H$ (Efficiency is 100%).
- This work is now supplied to a refrigerator that removes heat in the amount of Q_L from the low temperature reservoir and rejects heat in the amount of $Q_H + Q_L$ to the high temp. reservoir.
- During this process, the high temp. reservoir receives a net amount of heat Q_L (the difference between $Q_H + Q_L$ and Q_H).

FIGURE 6–26
Proof that the violation of the Kelvin–Planck statement leads to the violation of the Clausius statement.

6.4 Refrigerators and heat Pumps (10)

- Thus, the combination of these two devices can be viewed as a **refrigerator** as shown, that transfers heat in an amount of Q_L from a cooler body to a warmer one without requiring any input from outside.



(*b*) The equivalent refrigerator

- This is a violation of the Clausius statement.
- Therefore, a violation of the the Kelvin-Planck statement results in the violation of the Clausius statement.
- It can also be shown in a similar manner that a violation of the Clausius statement leads to the violation of the Kelvin–Planck statement.
- Therefore, the Clausius and the Kelvin–Planck statements are two equivalent expressions of the second law of thermodynamics.

FIGURE 6–26
Proof that the violation of the Kelvin–Planck statement leads to the violation of the Clausius statement.

Exercise 3

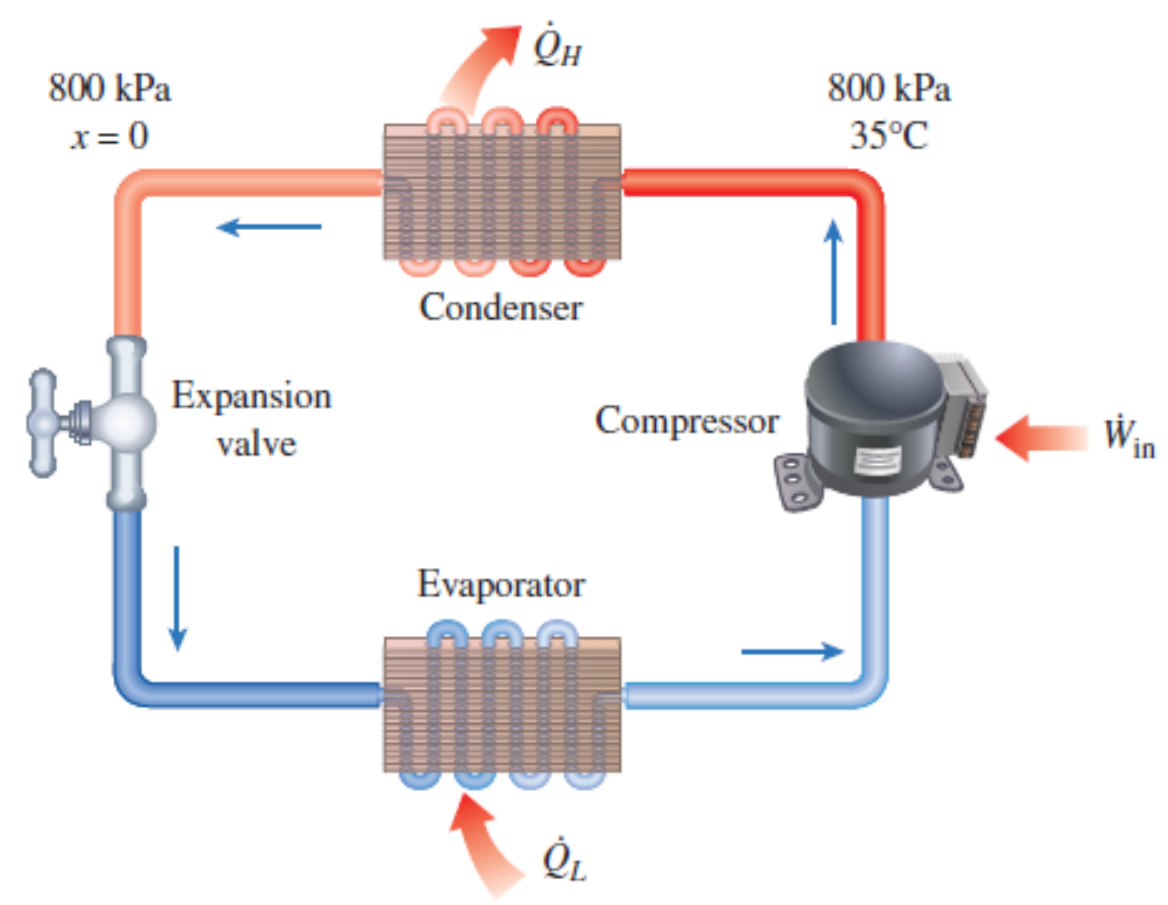
6–42 An air conditioner removes heat steadily from a house at a rate of 750 kJ/min while drawing electric power at a rate of 6 kW. Determine (*a*) the COP of this air conditioner and (*b*) the rate of heat transfer to the outside air. *Answers:* (*a*) 2.08, (*b*) 1110 kJ/min

Exercise 4

6–52 A heat pump used to heat a house runs about one-third of the time. The house is losing heat at an average rate of 22,000 kJ/h. If the COP of the heat pump is 2.8, determine the power the heat pump draws when running.

Exercise 5

6–55 Refrigerant-134a enters the condenser of a residential heat pump at 800 kPa and 35°C at a rate of 0.018 kg/s and leaves at 800 kPa as a saturated liquid. If the compressor consumes 1.2 kW of power, determine (*a*) the COP of the heat pump and (*b*) the rate of heat absorption from the outside air.



6.5 Perpetual-Motion Machines

- Any device that violates the first or second laws of thermodynamics is called a perpetual-motion machine.
  - If the device violates the first law (by creating energy), it is called a perpetual-motion machine of the first kind (PMM1).
  - If the device violates the second law, it is called a perpetual-motion machine of the second kind (PMM2)

6.5 Perpetual-Motion Machines (2)

Case 1

- The inventor claims that the plant :
  - heat the steam using part of electricity generated by the plant
  - will produce electricity indefinitely without requiring any energy input from the outside.
- The shaded area is continuously supplying energy to outside without receiving any energy, i.e. the system is creating energy
- This **violates the first law**.

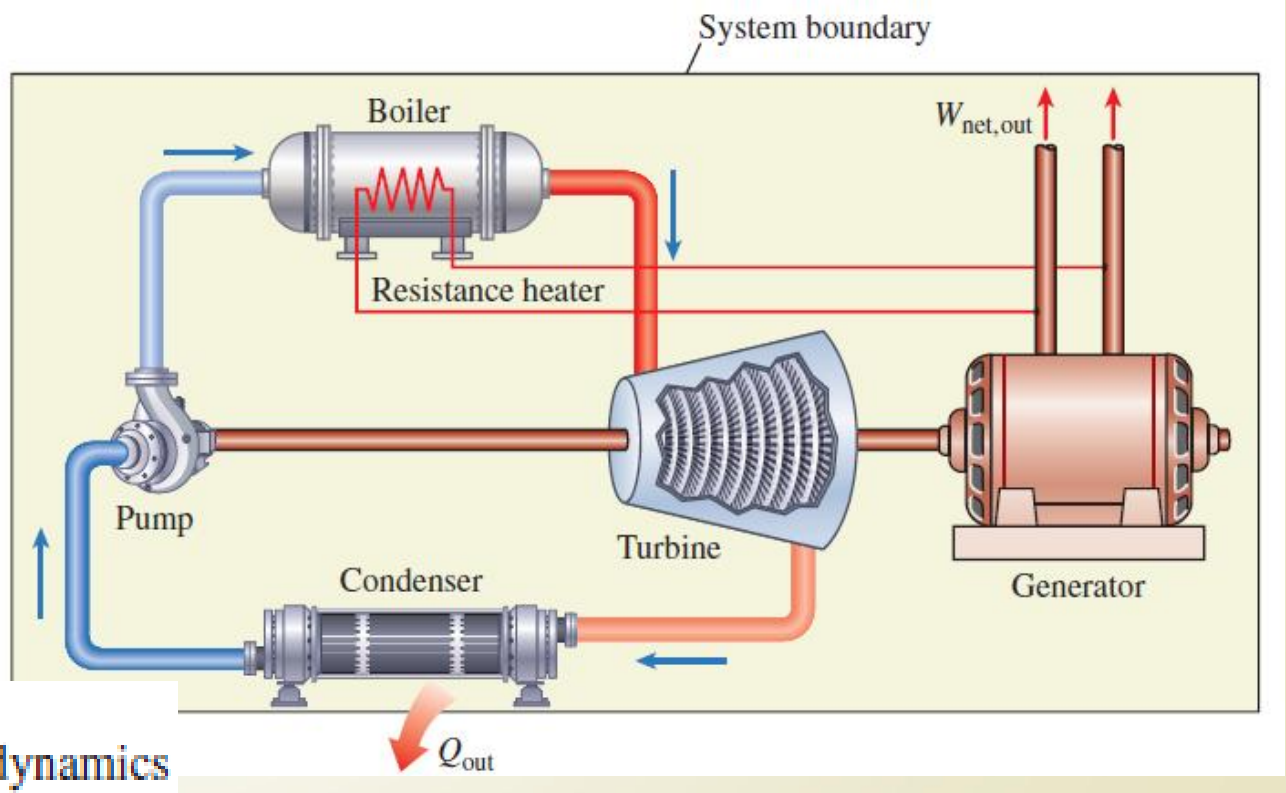


FIGURE 6–27
A perpetual-motion machine that violates the first law of thermodynamics (PMM1).

6.5 Perpetual-Motion Machines (3)

Case 2

- Realizing that more than one-half of the heat transferred to the steam in the furnace is discarded in the condenser to the environment,
  - an inventor suggests to get rid of the condenser and send the steam to the pump as soon as it leaves the turbine.
  - Thus, all the heat transferred to the steam will be converted to work (100% efficiency)
- But the device works on a cycle while exchanging heat with a single reservoir only.
- This satisfies the first law but **violates the second law**.

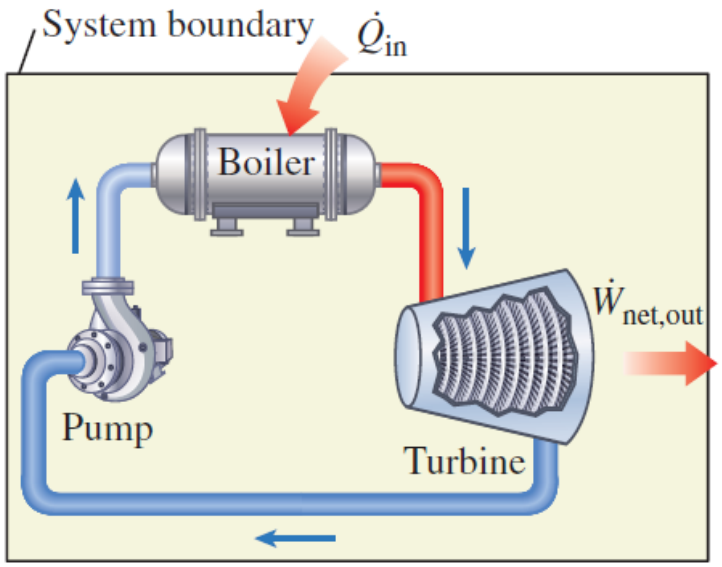


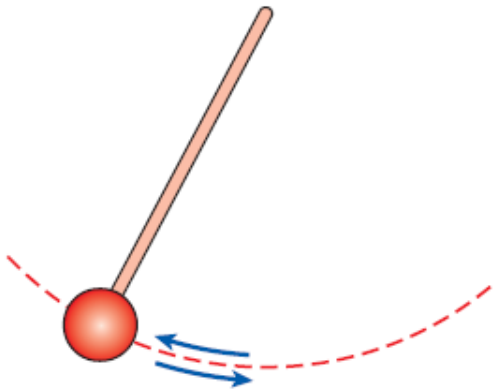
FIGURE 6–28
A perpetual-motion machine that violates the second law of thermodynamics (PMM2).

6.6 Reversible And Irreversible Processes

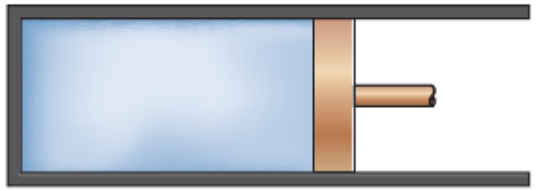
- Reversible process: A process that can be reversed without leaving any trace on the surroundings.
  - both the system and the surroundings are returned to their initial states at the end of the reverse process.
  - This is possible only if the net heat and net work exchange between the system and the surroundings is zero for the combined (original and reverse) process.
- Irreversible process: A process that is not reversible.
- Any system can be restored to its initial state following a process, regardless of whether the process is reversible or irreversible.
  - for reversible processes, this restoration is made without leaving any net change on the surroundings,
  - for irreversible processes, the surroundings usually do some work on the system and therefore do not return to their original state.

6.6 Reversible And Irreversible Processes (2)

- All the processes occurring in nature are irreversible.
- Why are we interested in reversible processes?
  - they are easy to analyze
  - they serve as idealized models (theoretical limits) to which actual processes can be compared.
- Some processes are more irreversible than others.
- The concept of reversible processes leads to the definition of **the second-law efficiency for actual processes**
  - It is the degree of approximation to the corresponding reversible processes.
  - Can compare the performance of different devices designed to do the same task on the basis of their efficiencies.
  - The better the design, the lower the irreversibilities and the higher the second-law efficiency.



(*a*) Frictionless pendulum



(*b*) Quasi-equilibrium expansion and compression of a gas

FIGURE 6–29
Two familiar reversible processes

6.6 Reversible And Irreversible Processes (3)

Irreversibilities

- The factors that cause a process to be irreversible are called irreversibilities.
  - E.g. friction, unrestrained expansion, mixing of two fluids, heat transfer across a finite temperature difference, electric resistance, and chemical reactions.
  - The presence of any of these effects renders a process irreversible.
- **Friction**
  - When two bodies in contact are forced to move relative to each other, a friction force that opposes the motion develops at the interface of these two bodies, and some work is needed to overcome this friction force.
  - The energy supplied as work is eventually converted to heat during the process
  - When the direction of the motion is reversed, the bodies are restored to their original position, but heat is not converted back to work.
  - Instead, more of the work is converted to heat
  - The larger the friction forces involved, the more irreversible the process is.

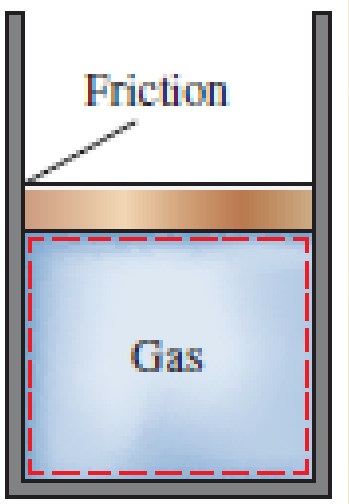
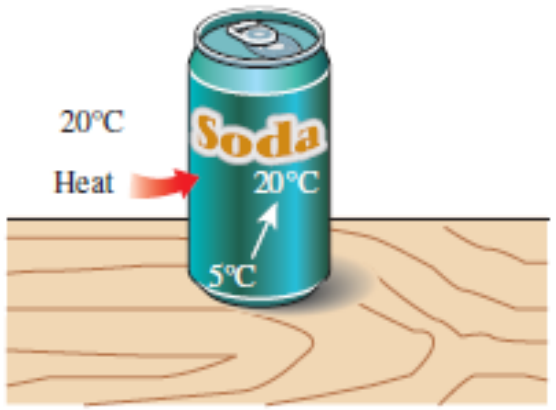


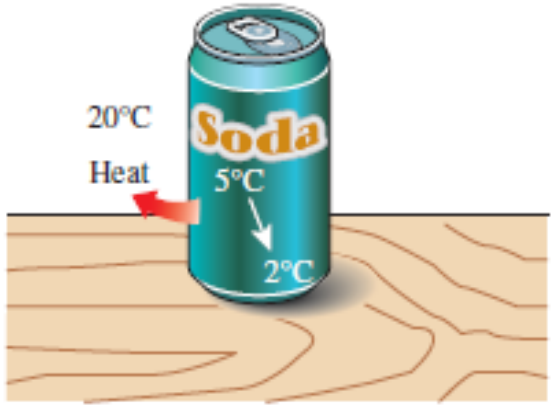
FIGURE 6–31
Friction renders a process irreversible.

6.6 Reversible And Irreversible Processes (4)

- **Heat transfer through a finite temperature difference**
  - Consider a can of cold soda left in a warm room
  - Heat is transferred from the warmer room air to the cooler soda.
  - The only way this process can be reversed and the soda restored to its original temperature is to provide refrigeration, which requires some work input.



(*a*) An irreversible heat transfer process



(*b*) An impossible heat transfer process

FIGURE 6–33
(*a*) Heat transfer through a temperature difference is irreversible, and (*b*) the reverse process is impossible.

6.6 Reversible And Irreversible Processes (3)

Internally and Externally Reversible Processes

- A typical process involves interactions between a system and its surroundings,
- A reversible process involves no irreversibilities associated with either of them.
- **Internally reversible process**: If no irreversibilities occur within the boundaries of the system during the process.
- **Externally reversible**: If no irreversibilities occur outside the system boundaries.
- **Totally reversible process**: involves no irreversibilities within the system or its surroundings.

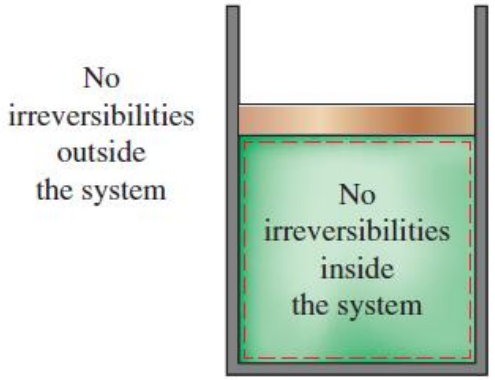


FIGURE 6–34
A reversible process involves no internal and external irreversibilities.

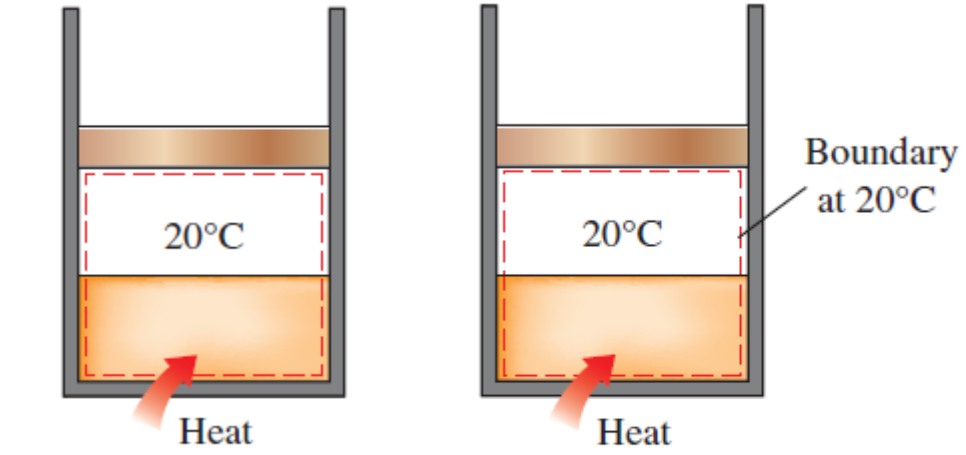
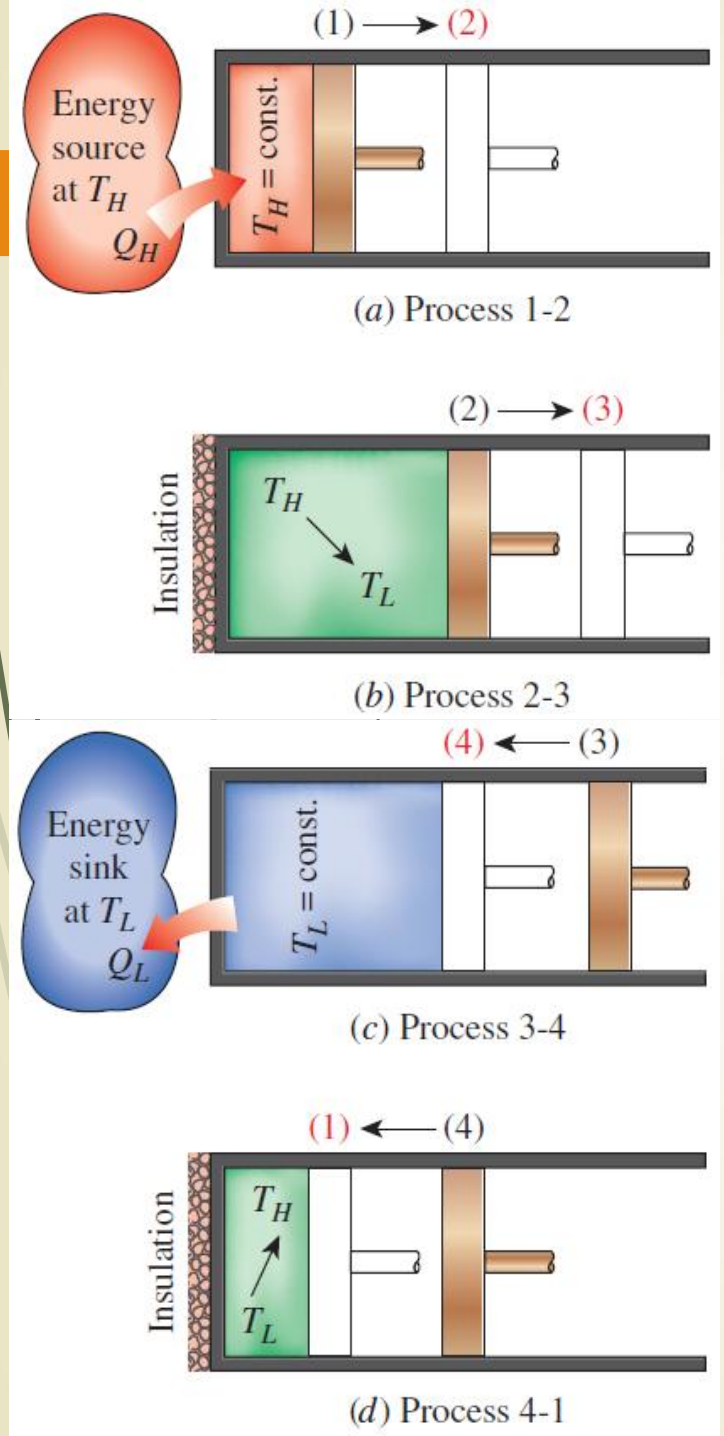


FIGURE 6–35
Totally and internally reversible heat transfer processes.

6.7 The Carnot Cycle

- Heat engines are cyclic devices and the working fluid returns to its initial state at the end of each cycle.
- The efficiency of a heat-engine cycle greatly depends on the net work (cycle efficiency), which is maximized by using processes that require the least amount of work and deliver the most, that is, by using reversible processes.
- Reversible cannot be achieved in practice, but it provide the upper limits on the performance of real cycles.
- Carnot was the first to introduce the concept of cyclic operation and devised a reversible cycle that is composed of four reversible processes, two isothermal and two adiabatic.



6.7 The Carnot Cycle (2)

- **Reversible Isothermal Expansion** (Process 1-2, T_H = const.) heat addition at high temperature, $T_H > T_L$ to the working fluid in a piston--cylinder device which does some boundary work.
- **Reversible Adiabatic Expansion** (Process 2-3) during which the system does work as the working fluid temperature decreases from T_H to T_L.
- **Reversible Isothermal Compression** (Process 3-4, T_L = const.) The system is brought in contact with a heat reservoir at $T_L < T_H$ and a reversible isothermal heat exchange takes place while work of compression is done on the system.
- **Reversible Adiabatic Compression** (Process 4-1) during this process the working fluid temperature increases from T_L to T_H.

6.7 The Carnot Cycle (3)

- In a *P-V* diagram the area under the process curve represents the boundary work
  - The area under curve 1-2-3 – the work done by the gas during the expansion part of the cycle
  - The area under curve 3-4-1 – the work done on the gas during the compression part of the cycle.
  - The area enclosed by the path of the cycle (area 1-2-3-4-1) is the difference between these two and represents the **net work done during the cycle**.

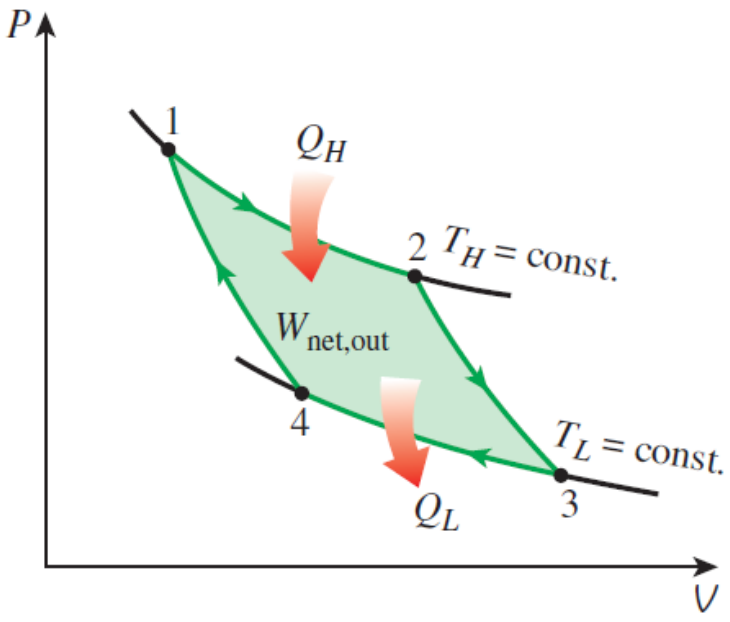


FIGURE 6–37
P-V diagram of the Carnot cycle.

6.7 The Carnot Cycle (4)

The Reversed Carnot Cycle

- The Carnot heat-engine cycle is a totally reversible cycle.
- Therefore, all the processes that compromise it can be reversed, in which case it becomes the **Carnot refrigeration cycle**.
- The *P-V* diagram of the reversed Carnot cycle is the same as the one given for Carnot cycle, except that the directions of the processes are reversed.
  - Heat in the amount of Q_L is absorbed from the low-temperature reservoir,
  - Heat in the amount of Q_H is rejected to a high-temperature reservoir, and
  - work input of $W_{net,in}$ is required

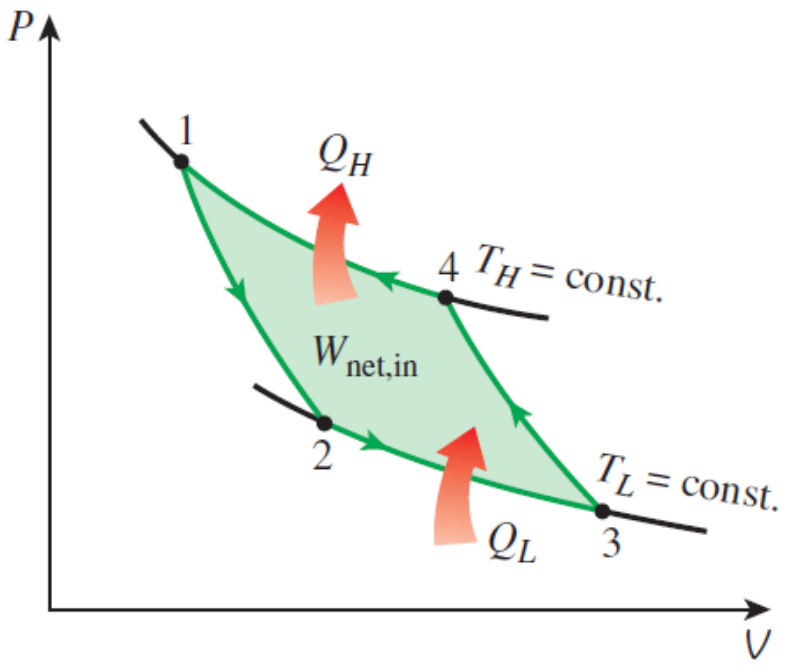


FIGURE 6–38
P-V diagram of the reversed Carnot cycle.

6.8 The Carnot Principles

- Two conclusions pertain to the thermal efficiency of reversible and irreversible (i.e., actual) heat engines, and they are known as the Carnot principles:

  1. The efficiency of an irreversible heat engine is always less than the efficiency of a reversible one operating between the same two reservoirs.

     $$\eta_{th} < \eta_{th,\,Carnot}$$

  2. The efficiencies of all reversible heat engines operating between same two constant temperature heat reservoirs have the same efficiency.

- For reversible process $$\left(\frac{Q_H}{Q_L}\right)_{rev} = \frac{T_H}{T_L}$$

  which indicate that the temperature ratios depend on the ratios of heat transfer between a **reversible** heat engine and the reservoirs, thus **independent of the physical properties of any substance**.

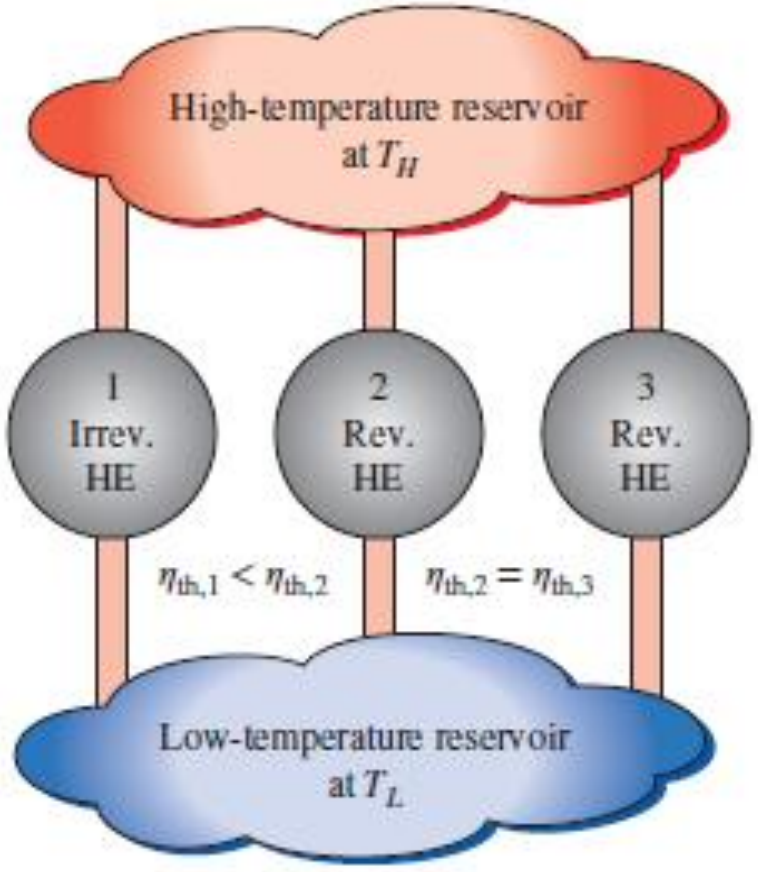


FIGURE 6–39
The Carnot principles.

6.10 The Carnot Heat Engine

- The hypothetical heat engine (HE) that operates on the reversible Carnot cycle is called the Carnot heat engine.
- The thermal efficiency of any heat engine reversible or irreversible,

$$\eta_{th} = 1 - \frac{Q_L}{Q_H}$$

- Then the efficiency of a Carnot engine (or any reversible heat engine)

$$\eta_{th,\,rev} = 1 - \frac{T_L}{T_H} \qquad \frac{Q_H}{Q_L} = \frac{T_H}{T_L}$$

- This is referred to as the **Carnot efficiency**.
- This is **the highest efficiency a heat engine** operating between the two thermal energy reservoirs at temperatures T_L and T_H can possibly have.
- **Note that T_L and T_H are absolute temperatures (Kelvin).**

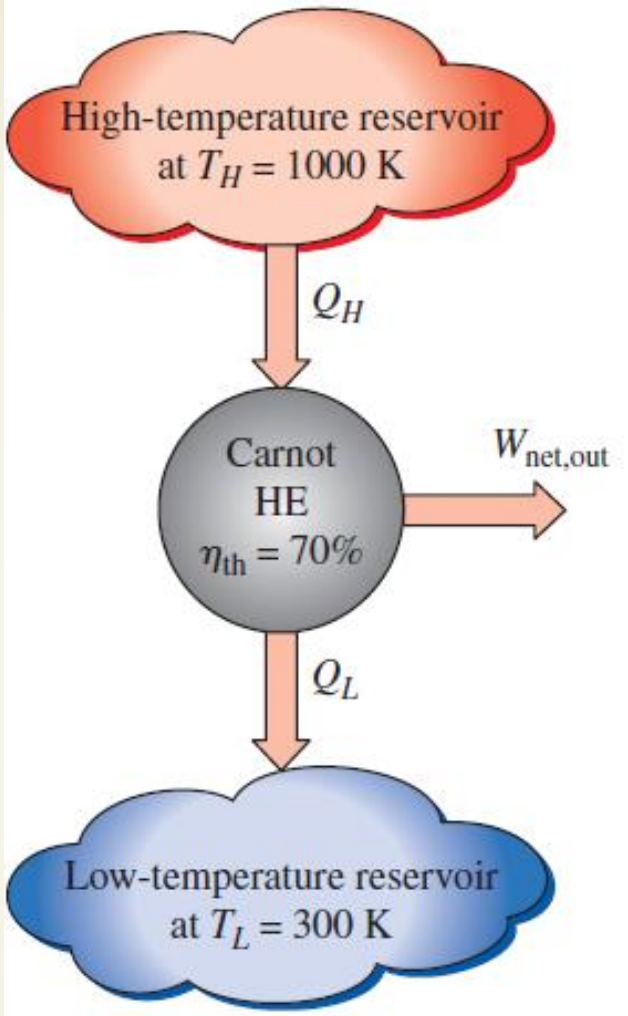


FIGURE 6–45
The Carnot heat engine is the most efficient of all heat engines operating between the same high- and low-temperature reservoirs.

6.10 The Carnot Heat Engine (2)

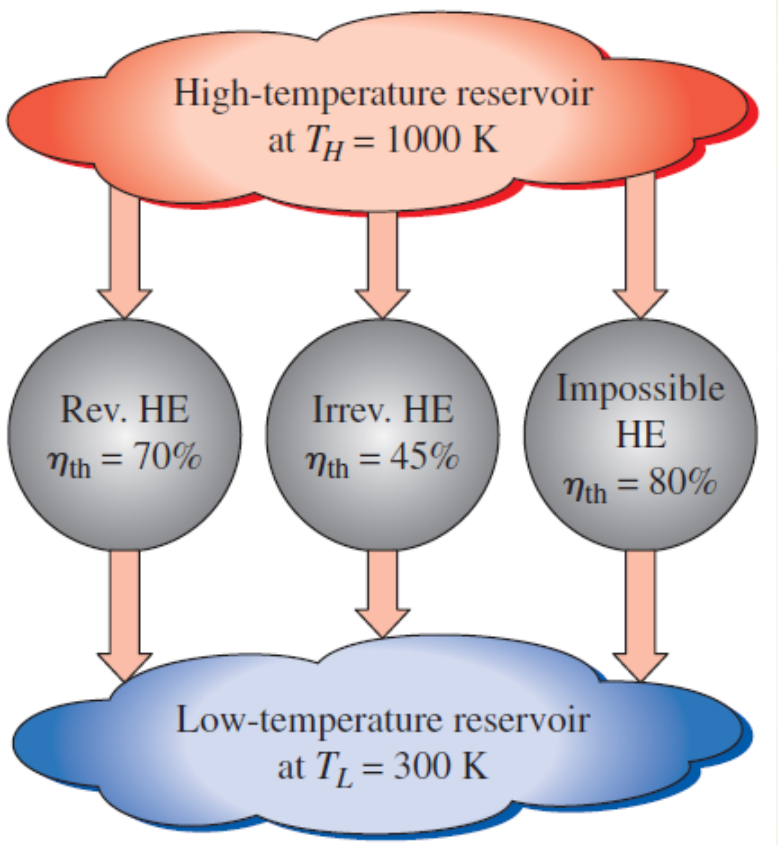


FIGURE 6–46
No heat engine can have a higher efficiency than a reversible heat engine operating between the same high- and low-temperature reservoirs.

- All irreversible (actual) HE operating between the same temperature limits (T_L and T_H) have lower efficiency.
- An actual HE cannot reach this maximum theoretical efficiency value since it is **impossible** to **completely eliminate** all the irreversibilities in actual cycle.

$$\eta_{th} \begin{cases} < \eta_{th,rev} & \text{irreversible heat engine} \\ = \eta_{th,rev} & \text{reversible heat engine} \\ > \eta_{th,rev} & \text{impossible heat engine} \end{cases}$$

- The thermal efficiency of actual HE can be maximized by supplying heat to the engine at the highest possible temperature and rejecting heat from the engine at the lowest possible temperature

EXAMPLE 6–5 Analysis of a Carnot Heat Engine

A Carnot heat engine, shown in Fig. 6–47, receives 500 kJ of heat per cycle from a high-temperature source at 652°C and rejects heat to a low-temperature sink at 30°C. Determine (*a*) the thermal efficiency of this Carnot engine and (*b*) the amount of heat rejected to the sink per cycle.

SOLUTION The heat supplied to a Carnot heat engine is given. The thermal efficiency and the heat rejected are to be determined.

Analysis (*a*) The Carnot heat engine is a reversible heat engine, and so its efficiency can be determined from Eq. 6–18 to be

$$\eta_{\text{th,rev}} = 1 - \frac{T_L}{T_H} = 1 - \frac{(30 + 273)\ \text{K}}{(652 + 273)\ \text{K}} = \mathbf{0.672}$$

That is, this Carnot heat engine converts 67.2 percent of the heat it receives to work.

(*b*) The amount of heat rejected Q_L by this reversible heat engine is easily determined from Eq. 6–16 to be

$$Q_{L,\text{rev}} = \frac{T_L}{T_H} Q_{H,\text{rev}} = \frac{(30 + 273)\ \text{K}}{(652 + 273)\ \text{K}} (500\ \text{kJ}) = \mathbf{164\ kJ}$$

Discussion Note that this Carnot heat engine rejects to a low-temperature sink 164 kJ of the 500 kJ of heat it receives during each cycle.

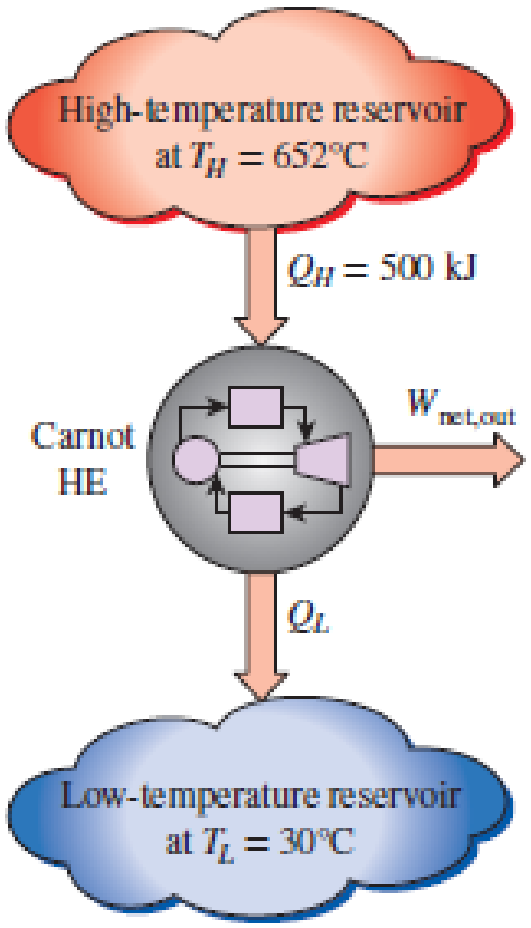
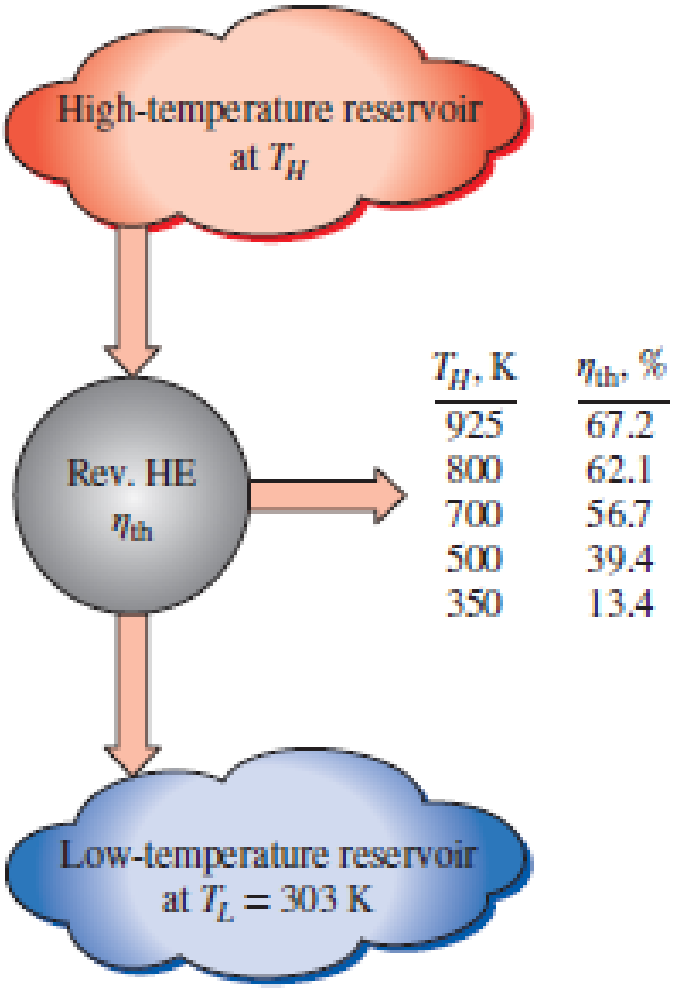


FIGURE 6–47
Schematic for Example 6–5.

6.10 The Carnot Heat Engine (3)



T_H, K	η_{th}, %
925	67.2
800	62.1
700	56.7
500	39.4
350	13.4

FIGURE 6–48
The fraction of heat that can be converted to work as a function of source temperature (for $T_L = 303$ K).

The Quality of Energy

- Figure 6-48 shows that the thermal efficiency varies with the source temperature when the sink temperature is held constant.
  - the thermal efficiency decreases as the source temperature is lowered
  - more of the high-temperature thermal energy can be converted to work.
- These efficiency values show that energy has **quality** as well as quantity.
- It can be stated that:

  The higher the temperature, the higher the **quality** of the energy
- Work is a more valuable form of energy than heat since 100% of work can be converted to heat, but only a fraction of heat can be converted to work.

Exercise 5

6–74 From a work-production perspective, which is more valuable: (*a*) thermal energy reservoirs at 675 K and 325 K or (*b*) thermal energy reservoirs at 625 K and 275 K?

Exercise 6

6–81 A heat engine receives heat from a heat source at 1200°C and has a thermal efficiency of 40 percent. The heat engine does maximum work equal to 500 kJ. Determine the heat supplied to the heat engine by the heat source, the heat rejected to the heat sink, and the temperature of the heat sink.

6.11 The Carnot Refrigerator and Heat Pump

- A refrigerator or a heat pump that operates on the reversed Carnot cycle is called a **Carnot refrigerator**, or a **Carnot heat pump**.
- Recall that COP for refrigerator or heat pump generally is given by

$$\mathrm{COP_R} = \frac{1}{Q_H/Q_L - 1} \quad \text{and} \quad \mathrm{COP_{HP}} = \frac{1}{1 - Q_L/Q_H}$$

- Then the COP for reversible (Carnot) refrigerators and heat pumps become

$$\mathrm{COP_{R,rev}} = \frac{1}{T_H/T_L - 1} \quad \text{and} \quad \mathrm{COP_{HP,rev}} = \frac{1}{1 - T_L/T_H}$$

- The COP of a reversible refrigerator or heat pump is the **maximum theoretical value** for the specified temperature limits.
- Actual refrigerators or heat pumps may approach these values as their designs are improved, but they can never reach them.
- The COPs of both the refrigerators and the heat pumps decrease as T_L decreases.
  - it requires more work to absorb heat from lower-temperature media.

$$\mathrm{COP_R} \begin{cases} < \mathrm{COP_{R,rev}} & \text{irreversible refrigerator} \\ = \mathrm{COP_{R,rev}} & \text{reversible refrigerator} \\ > \mathrm{COP_{R,rev}} & \text{impossible refrigerator} \end{cases}$$

EXAMPLE 6–6 A Carnot Refrigeration Cycle Operating in the Saturation Dome

A Carnot refrigeration cycle is executed in a closed system in the saturated liquid–vapor mixture region using 0.8 kg of refrigerant-134a as the working fluid (Fig. 6–51). The maximum and the minimum temperatures in the cycle are 20 and –8°C, respectively. It is known that the refrigerant is saturated liquid at the end of the heat rejection process, and the net work input to the cycle is 15 kJ. Determine the fraction of the mass of the refrigerant that vaporizes during the heat addition process and the pressure at the end of the heat rejection process.

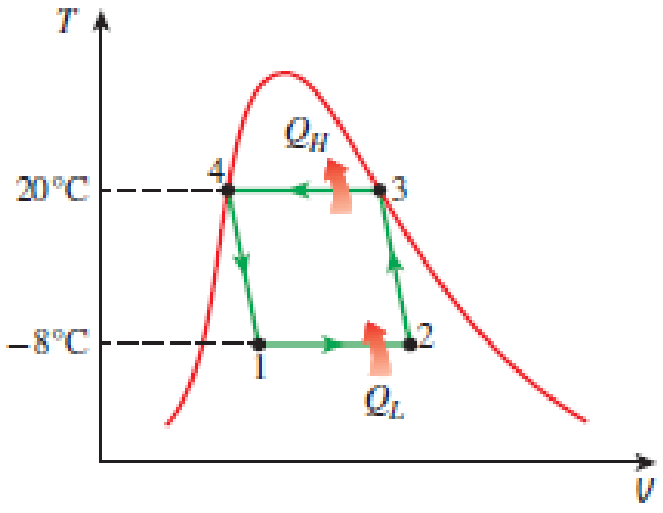


FIGURE 6–51
Schematic for Example 6–6.

SOLUTION A Carnot refrigeration cycle is executed in a closed system. The mass fraction of the refrigerant that vaporizes during the heat addition process and the pressure at the end of the heat rejection process are to be determined.

Assumptions The refrigerator operates on the ideal Carnot cycle.

Analysis Knowing the high and low temperatures, the coefficient of performance of the cycle is

$$\mathrm{COP}_R = \frac{1}{T_H/T_L - 1} = \frac{1}{(20 + 273\ \mathrm{K})/(-8 + 273\ \mathrm{K}) - 1} = 9.464$$

The amount of cooling is determined from the definition of the coefficient of performance to be

$$Q_L = \text{COP}_R \times W_{in} = (9.464)(15\text{ kJ}) = 142\text{ kJ}$$

The enthalpy of vaporization of R-134a at –8°C is $h_{fg} = 204.59$ kJ/kg (Table A–11). Then the amount of refrigerant that vaporizes during heat absorption becomes

$$Q_L = m_{evap} h_{fg\,@\,-8°C} \rightarrow m_{evap} = \frac{142\text{ kJ}}{204.59\text{ kJ/kg}} = 0.694\text{ kg}$$

Therefore, the fraction of mass that vaporized during the heat addition process to the refrigerant is

$$\text{Mass fraction} = \frac{m_{evap}}{m_{total}} = \frac{0.694\text{ kg}}{0.8\text{ kg}} = \mathbf{0.868}\text{ (or 86.8 percent)}$$

The pressure at the end of the heat rejection process is simply the saturation pressure at heat rejection temperature,

$$P_4 = P_{sat\,@\,20°C} = \mathbf{572.1\text{ kPa}}$$

Discussion The reversed Carnot cycle is an idealized refrigeration cycle, thus it cannot be achieved in practice. Practical refrigeration cycles are analyzed in Chap. 11.

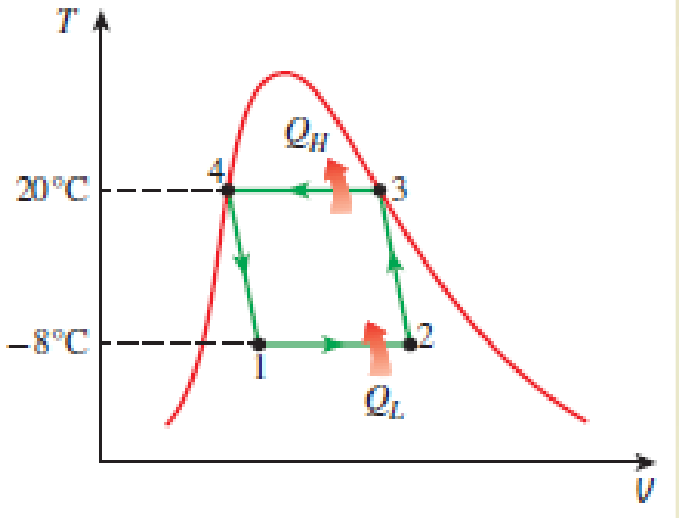


FIGURE 6–51
Schematic for Example 6–6.

EXAMPLE 6–7 Heating a House with a Carnot Heat Pump

A heat pump is to be used to heat a house during the winter, as shown in Fig. 6–52. The house is to be maintained at 21°C at all times. The house is estimated to be losing heat at a rate of 135,000 kJ/h when the outside temperature drops to –5°C. Determine the minimum power required to drive this heat pump.

SOLUTION A heat pump maintains a house at a constant temperature. The required minimum power input to the heat pump is to be determined.

Assumptions Steady operating conditions exist.

Analysis The heat pump must supply heat to the house at a rate of $\dot{Q}_H = 135{,}000\ \text{kJ/h} = 37.5\ \text{kW}$. The power requirements are minimum when a reversible heat pump is used to do the job. The COP of a reversible heat pump operating between the house and the outside air is

$$\text{COP}_{\text{HP,rev}} = \frac{1}{1 - T_L/T_H} = \frac{1}{1 - (-5 + 273\ \text{K})/(21 + 273\ \text{K})} = 11.3$$

Then, the required power input to this reversible heat pump becomes

$$\dot{W}_{\text{net,in}} = \frac{\dot{Q}_H}{\text{COP}_{\text{HP}}} = \frac{37.5\ \text{kW}}{11.3} = \mathbf{3.32\ kW}$$

Discussion This reversible heat pump can meet the heating requirements of this house by consuming electric power at a rate of 3.32 kW only. If this house were to be heated by electric resistance heaters instead, the power consumption would jump up 11.3 times to 37.5 kW. This is because in resistance heaters the electric energy is converted to heat at a one-to-one ratio. With a heat pump, however, energy is absorbed from the outside and carried to the inside using a refrigeration cycle that consumes only 3.32 kW. Notice that the heat pump does not create energy. It merely transports it from one medium (the cold outdoors) to another (the warm indoors).

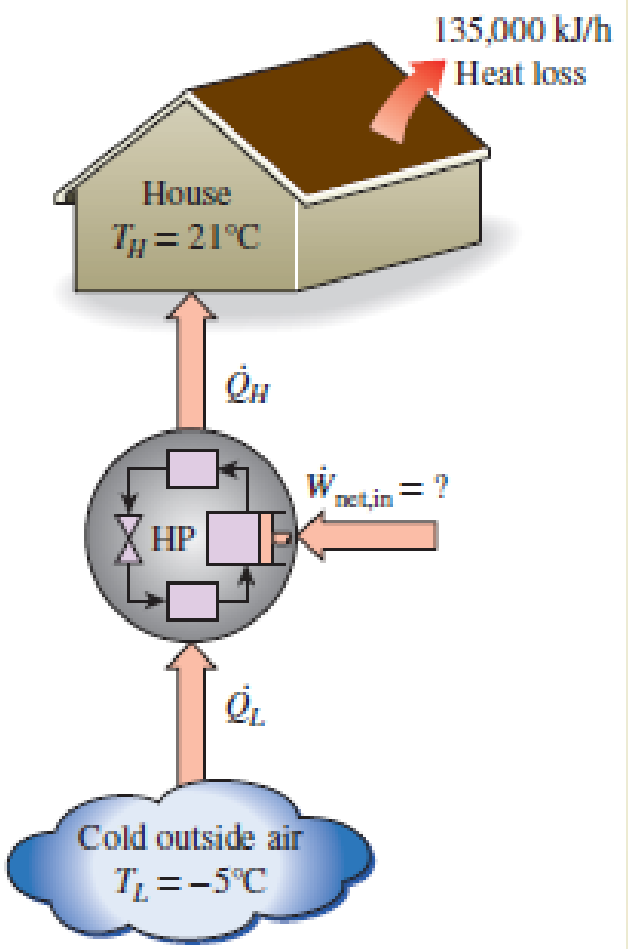
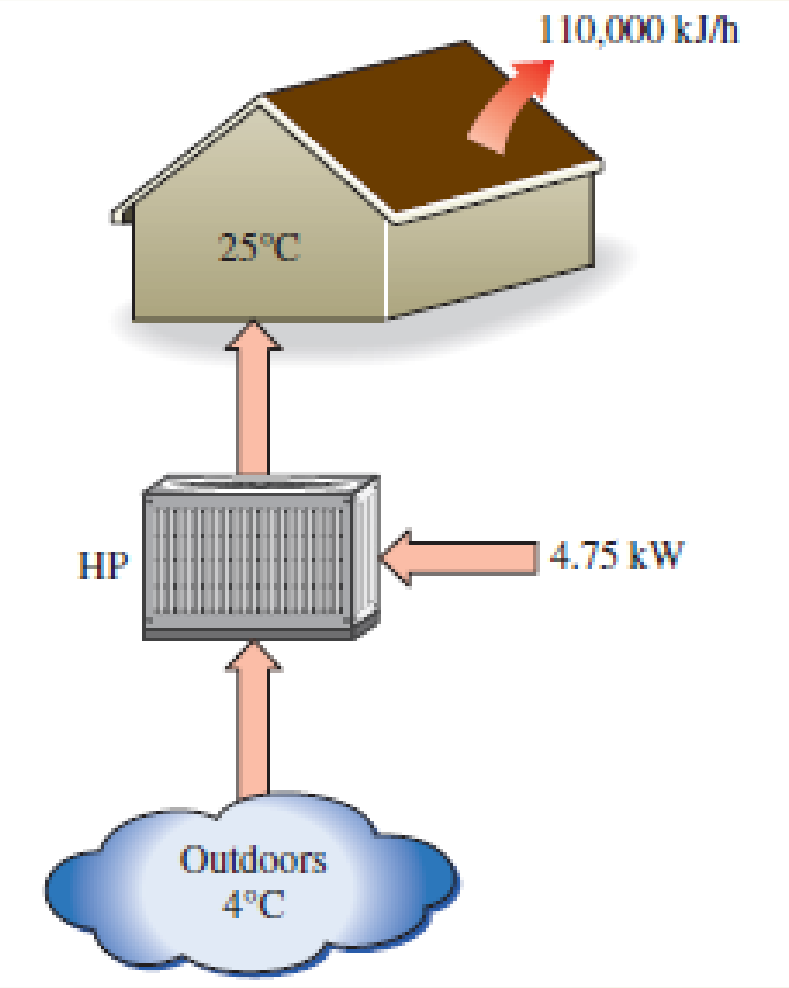


FIGURE 6–52
Schematic for Example 6–7.

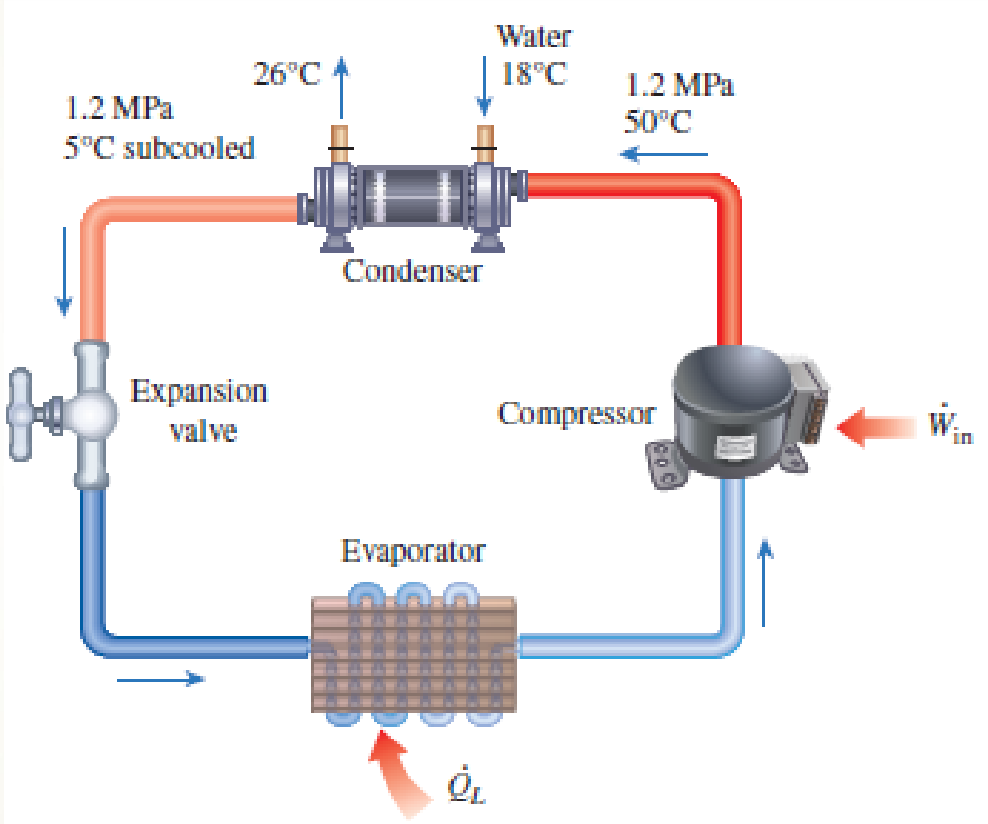
Exercise 7

6–97 A heat pump is used to maintain a house at 25°C by extracting heat from the outside air on a day when the outside air temperature is 4°C. The house is estimated to lose heat at a rate of 110,000 kJ/h, and the heat pump consumes 4.75 kW of electric power when running. Is this heat pump powerful enough to do the job?



Exercise 8

6–101 A commercial refrigerator with refrigerant-134a as the working fluid is used to keep the refrigerated space at −35°C by rejecting waste heat to cooling water that enters the condenser at 18°C at a rate of 0.25 kg/s and leaves at 26°C. The refrigerant enters the condenser at 1.2 MPa and 50°C and leaves at the same pressure subcooled by 5°C. If the compressor consumes 3.3 kW of power, determine (*a*) the mass flow rate of the refrigerant, (*b*) the refrigeration load, (*c*) the COP, and (*d*) the minimum power input to the compressor for the same refrigeration load.



Exercise 9

6–108 An air-conditioner with refrigerant-134a as the working fluid is used to keep a room at 23°C by rejecting the waste heat to the outdoor air at 34°C. The room gains heat through the walls and the windows at a rate of 250 kJ/min while the heat generated by the computer, TV, and lights amounts to 900 W. The refrigerant enters the compressor at 400 kPa as a saturated vapor at a rate of 80 L/min and leaves at 1200 kPa and 70°C. Determine (*a*) the actual COP, (*b*) the maximum COP, and (*c*) the minimum volume flow rate of the refrigerant at the compressor inlet for the same compressor inlet and exit conditions. *Answers:* (*a*) 4.33, (*b*) 26.9, (*c*) 12.9 L/min

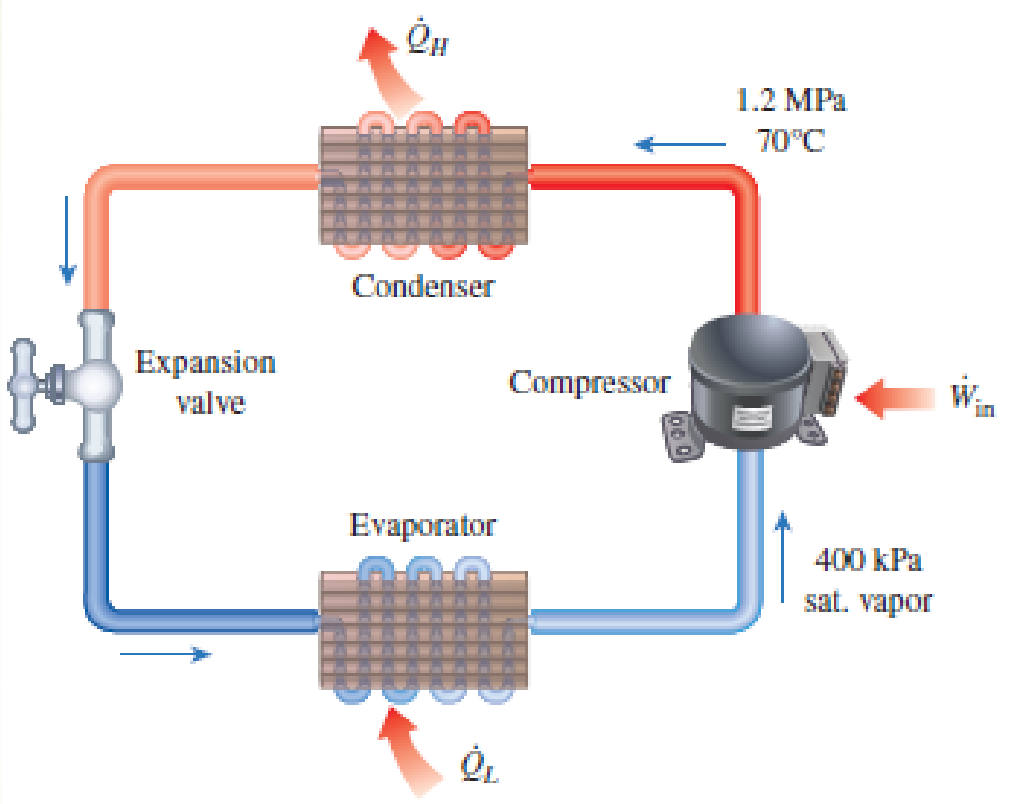


FIGURE P6–108

Thank You